

Thesis

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Operationalizing Low Code/No Code

Exploratory Interviews with Hungarian LCNC Professionals and a Microsoft Power Platform Case-Study

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1 Paper Introduction

In the age of AI and cloud computing, technology's role in society is expanding rapidly. As public discourse often emphasizes the impact of AI on business strategy and digital transformation, several challenges might surface, such as the potential devaluation of routine administrative work and the widening digital skills gap in the workforce. While businesses often "over-hype" the role of AI, a parallel class of cloud technologies with a comparable value proposition seems less prominent. LCNC technologies propose the democratization of software creation by enabling non-IT professionals to build and customize digital solutions without the need for deep programming expertise and large investments (IBM, n.d.b).

While LCNC technologies became widely available in the second half of the 2010s, the underlying impulse to optimize workflows and create digital solutions by non-IT practitioners has a longer history. As digital tools have become increasingly accessible, professionals with digital aptitudes have sought to utilize them in a way that is beneficial for themselves and their respective environments. This impulse was defined as citizen development (CD). While the concept of CD is much broader, the two terms (CD and LCNC) have become intertwined. LCNC platforms are powerful and easy-to-use tools for business professionals to create added value for their customers and make their own or their colleagues' lives easier (Binzer & Winkler, 2022).

Despite the increasing maturity of LCNC platforms, organizations often struggle to realize their underlying value and potential. In operationalizing CD and LCNC at scale, several obstructions can emerge at both the organizational and individual levels. This thesis is meant to contextualize these challenges and provide practical insights to address them. The importance of these conclusions lies not only in enabling citizen developers but also in understanding broader implications. As AI adoption is becoming an explicit competitive factor and a growing amount of work is being performed through dispersed technologies in cross-functional domains, understanding the state of LCNC today might also provide valuable lessons for all similar technologies in the future.

The aim of this thesis is to develop a practical, empirically supported understanding of how organizations can leverage the benefits of utilizing LCNC platforms through CD initiatives and to provide actionable insights for future CD projects.

RQ1. How does CD contribute to organizational strategy and digital transformation?

RQ2. What organizational and individual factors enable an efficient CD strategy?

To answer these questions, the thesis derives discussable topics, implications and questions from the literature review, that provide the framework for the empirical sections.

In the empirical sections the thesis introduces the findings of two-method qualitative research, utilizing a case study and qualitative interviews with LCNC professionals.

2 Literature Review

2.1 Digital Transformation (DT) in Modern Organizations

In recent years, digital transformation has emerged as a central theme for organizations seeking to remain competitive in increasingly dynamic markets and meet evolving customer expectations. Traditionally, information technology (IT) was regarded as an enabling function that should be aligned with, rather than integral to, business strategy (Bharadwaj et al., 2013; Henderson & Venkatraman, 1993). However, the past two decades have demonstrated that digital technologies can fundamentally transform industries, reshaping value creation and competitive dynamics, e.g., streaming reshaped media consumption.. According to Bharadwaj et al. (2013), digital technologies have shifted from supporting business operations to forming the core of digital business strategy—blurring the traditional boundaries between IT and business. Furthermore, the declining cost, increasing availability, and reduced hardware requirements of cloud computing have lowered entry barriers, enabling smaller players and new market entrants to leverage digital tools as a source of competitive advantage against larger enterprises (Marston et al., 2011).

Bharadwaj et al. (2013) use a new conceptualization of IT strategy by elevating it from a supporting functional role to an integral component of business strategy. They introduce the notion of digital business strategy, defined as “organizational strategy formulated and executed by leveraging digital resources to create differential value” (Bharadwaj et al. 2013, p. 472). **This reframing implies that IT is no longer confined to a back-office or in an enabling function’s role but instead becomes a central driver of competitive advantage and innovation.**

The authors identify three central elements of this definition.

- Emphasizing that digital capabilities exist in all parts of the organization, including marketing, sales, logistics. Consequently, IT departments should move from full ownership of digital assets to a shared ownership model, maintaining oversight of security and compliance while empowering business units to co-own digital initiatives.
- It moves beyond a technology- or product-oriented perspective of IT strategy towards a resource-based view (RBV), recognizing digital resources (e.g., data, platforms, analytics capabilities) as strategic assets that not just support business processes, but actively create differentiating value for the business.
- It explicitly links digital strategy to performance outcomes, extending its scope beyond operational efficiency to include innovation, customer experience, and market positioning.

With the advancement of IT from a supporting function to a central component of digital business strategy, it is also important to consider the transformational impact of digital technologies on the organization. While the most notable examples of industry-wide disruption are often startups or highly technology-focused companies, a multi-year study by MIT Sloan School of Management and Capgemini Consulting (MIT Center for Digital Business & Capgemini Consulting, 2011) demonstrates that established organizations can also successfully leverage digital technologies to transform business processes, customer experiences, and value creation models. The key findings of this research are:

- The pace and success of digital transformation highly depends on the company’s internal and external circumstances, such as organizational culture, skills, market- and customer-pressure.
- Implementing new technologies is often not the way organizations can benefit from DT, rather than re-envisioning business processes and models in a way they are enhanced by technology. This also resonates with Bharadwaj et al. in distributing digital resources and capabilities across functions and co-owning these resources with IT.
- The research also states, that successful digital innovation does not start bottom-up, but has to be driven from the top. However, this does not mean merely the adoption of certain digital technologies, rather than creating the internal demand, for such transformative resources.

It is important to note that, while the authors (MIT Center for Digital Business & Capgemini Consulting, 2011) do not discuss specific technologies, their research predates both the AI boom of the late 2010s and the first public definitions of LCNC platforms.

As organizations continue to evolve in response to the pressures and opportunities of digital transformation, a growing need has emerged for approaches that enable faster, more adaptive, and more inclusive innovation. **Traditional, centralized IT-led transformation models often struggle to keep pace with rapidly changing business demands, limited technical resources, and the need for iterative experimentation (Klotz et al., 2019).** In this context, LCNC platforms have gained prominence as technologies that lower the technical barriers to software development, allowing business professionals to contribute directly to digital solution creation. This shift has given rise to the concept of CD, where non-IT employees become active participants in digital innovation, marking a new phase in how organizations operationalize digital transformation (Gartner, 2021a).

2.2 LCNC and Citizen Development as Evolving Concepts

From a technical point of view, LCNC development platforms are seen as tools for the rapid development of (business) applications by utilizing predefined objects and logic, often featuring drag-and-drop and click-based functions with minimal (Low Code) or no need (No Code) for custom coding (IBM, n.d.b). In this concept, the use of LCNC platforms enable developers for rapid prototyping and utilizing hyper-agile methodologies, offering a way for enhanced feedback opportunities from the customer side (Carroll & Ajimati, 2024).

While several vendors explicitly separate Low Code and No Code development (Mendix, n.d.; Microsoft, n.d.; OutSystems, n.d.), most commercially available platforms have extended functionality to insert custom codes, blurring the line between the two. As both approaches share the goal of abstracting development from coding skills, reducing costs and accelerating delivery (Verma & Shree, 2025), this paper uses LCNC as an umbrella term.

Even though LCNC is a software development tool, its primary and potentially disruptive value lies in user enablement. In this sense LCNC development depends on visual platforms, abstracted from coding, that enable users with little to no coding experience to create functioning, value-adding solutions for themselves or other users. **This paper is going to discuss LCNC as a user enabler technology closely connected to Citizen Development (CD).**

Understanding the concept of CD helps us realize the value of user enablement. The idea behind this term, to develop software without coders can be traced back to 1982, when James Martin published his work "Application Development Without Programmers" (Martin, 1982), discussing the possibilities in 4th generational languages and Computer Assisted Software-Engineering tools. This early idea could be seen as both a testament for the early ideals of eDemocracy (e.g. ARPANET) and as the foreshadow of the CD initiatives of today. If we look back at how these technologies evolved, we can see that while some 4th generational languages still live on, they were not widely spread, apart from some technical areas, such as data engineering (SQL). ABAP is also a language that still has a role in business environments; however, this can be attributed to the emerging use of SAP and is not widely used in the development of new solutions. While 4th generational languages might not have had the impact as predicted, the idea of creating software and digital solutions by non-IT personnel was not forgotten (Verma & Shree, 2025). Before the wide availability of LCNC platforms, early "citizen developers" relied mostly on manual coded scripts and application integrated languages, such as Excel VBA codes. Even though this was and still is a highly efficient way of optimizing workflows, there are several limitations to these tools, such as relatively high entry-limit regarding skills and knowledge, limited availability for distribution. It is also worth to mention governance limitations and issues, related to the phenomena of "Shadow IT" (Goth, 2021).

Shadow IT is a general term, for applications and digital solutions used without the approval or knowledge of enterprise IT departments (IBM, n.d.c). According to research outlined by Cisco in 2016, 80% of end users rely on solutions considered shadow IT (Cisco, 2016). While Shadow IT can be seen as a risk for the organization, it is also important to note that in some cases these ungoverned solutions

were intended to unlock further operational efficiency and as such, were the precursors of Citizen Development.

By the second half of the 2010s we can observe a booming of enterprise LCNC platforms, leading to Gartner publishing its first magic quadrant explicitly for low code application platforms in 2019 (Gartner, 2019). In this report non-IT user enablement is part of the definition to classify these platforms, evaluating 18 vendors such as Microsoft, Salesforce, Oracle, ServiceNow and Appian.

In 2021 the Project Management Institute published the first vendor agnostic framework for citizen development (Project Management Institute, 2021). According to the definition of PMI, citizen development is a practice that allows non-technical employees, also known as "citizen developers", to create applications or automate processes using LCNC platforms.

By tracing this evolution, it becomes evident that **LCNC technologies addressed preexisting needs for user-driven application development and provided a scalable alternative to Shadow IT**—even before the AI revolution of the early 2020s. With the publication of the PMI Citizen Development® framework (Project Management Institute, 2021), the role of the citizen developer has been formally recognized and embedded within organizational practice.

2.3 Impact of CD in DT

While the previously discussed concept of digital business strategy emphasizes the distribution of digital resources across business functions, it also highlights a persistent gap between the growing digital innovation needs of business units and the limited capacity of traditional IT departments and professional developers (Hoogsteen & Borgman, 2022). **As organizations strive to innovate rapidly and adapt to dynamic markets, IT departments often struggle to meet escalating demands for digital solutions. This imbalance not only slows transformation initiatives but also limits the organization's ability to leverage digital opportunities proactively, positioning IT more as a reactive function than an enabler. Citizen Development (CD) further expands digital transformation (DT) capacity by empowering non-technical employees to actively participate in solution development.** This democratization of innovation diffuses digital capabilities across the enterprise, reducing dependence on traditional IT structures and fostering organizational agility (Carroll & Maher, 2023). Carroll & Maher further emphasize that the successful integration of CD depends on establishing clearly defined governance standards, coordinated and supervised by IT departments to ensure alignment and system integrity.

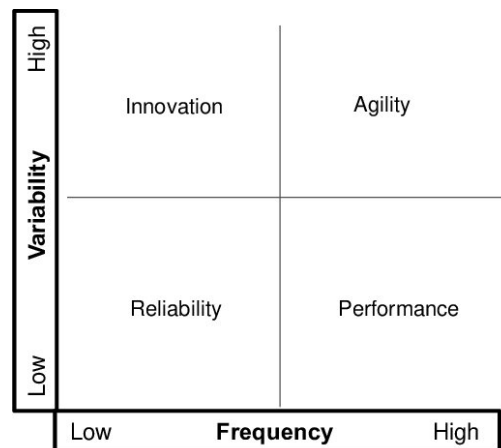
Regarding the use and availability of LCNC platforms – enabling technologies of CD, we can also see a shift towards higher adoption among companies. A 2022 KPMG survey of 715 organizations across the EMEA region found that only 15% of companies had not yet decided whether to adopt LCNC development platforms, while 22% had already made LCNC a core component of their software development strategy. These findings indicate a growing recognition among enterprises that LCNC technologies are not merely tools for faster development, but strategic enablers of digital transformation through wider employee participation and accelerated innovation cycles (KPMG International, 2023).

However, for organizations to fully realize the potential of CD, its implementation must be strategically aligned with existing process structures and governance models.

2.4 Organizational alignment of CD initiatives

In general, the role of a citizen developer is not a formal job title but a functional role that emerges organically within business units. Understanding this requires examining how CD can be aligned with an organization's broader process strategy. To analyze this alignment, this paper applies the BPM Context Matrix (vom Brocke et al., 2021), which categorizes business processes based on their variability and frequency.

The BPM context matrix categorizes processes by two axes, variability and frequency. Variability is dependent on how many external and internal factors a process has to align with, while frequency is used to measure the number of executions by a given time.



1. Figure BPM Context Matrix by vom Brocke et al. (2021)

Based on these two variables, four process classes are identified.

Low variability – Low frequency, **“Reliability” processes are defined by their uniform execution**, such as deriving a yearly financial statement from an ERP statement with the same outputs each year.

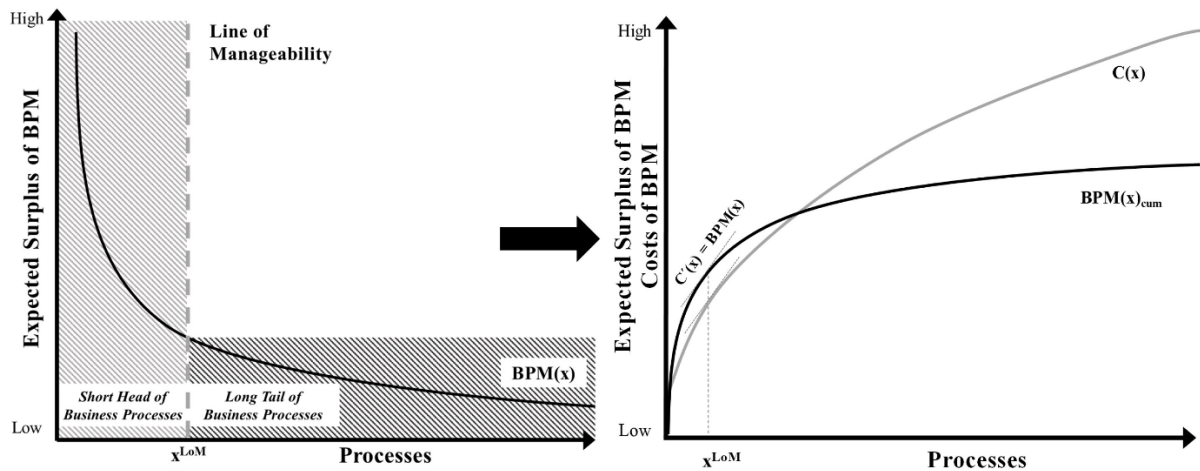
Low variability – High frequency, **“Performance” processes can be defined by similar execution and outputs to the Reliability processes, however, their frequency is higher**. A good example area for such processes is procurement or production. It is also worth mentioning that in some cases these processes are connected to or used by Reliability processes.

High variability – Low frequency, **“Innovation” processes are characterized by their varying outputs and execution**. These processes can create strategic value and determine the value proposal of an organization. This cluster requires a high-level of creativity and skill and is highly dependent on the human actors. A good example is R&D.

High variability – High frequency, **“Agility” processes are highly dependent on the knowledge of the actors, or more precisely on their area of expertise**. Often, these processes rely on improvisation or adaptability.

From a global perspective frequency and variability can highly differ from a local perspective. While most enterprises make significant investments in their global high frequency processes, on a local scale we can see that several tasks can fall into highly different quadrants in the matrix.

For example, if a company’s sales processes vary significantly across the countries where its subsidiaries operate, then from a global perspective they are high-variability, low-frequency. From a peripheral perspective, however, each subsidiary executes its own stable pattern frequently and with low variability, typically around a global core process. **This phenomenon is often referred to as the “long tail of work”**. From an enterprise perspective these long-tail processes are often not viewed as eligible for optimization as the resources needed to effectively manage these processes outweigh the impact, reducing the overall utility of BPM (Imgrund et al., 2017).



2. Figure The long tail of business processes by Imgrund et al (2017)

CD creates value by reducing the marginal cost $C(x)$ of automating and formalizing high-frequency, lower-level processes through end-user empowerment, typically with little to no additional IT spend. As local ownership increases, the enterprise BPM Context Matrix can be realized at lower organizational levels, raising the cumulative expected surplus of BPM ($BPM(x)_{cum}$). By simultaneously lowering BPM costs and increasing expected benefits, the Line of Manageability shifts rightward into the “long tail” of processes (Figure 2). This approach strongly connects to hyper-automation—a contemporary concept that aims to automate all automatable processes within an organization (IBM, n.d.-a). In doing so, organizations can rapidly extend automation across multiple processes without overburdening IT teams. At the same time, it fosters hyper-agility by empowering employees to quickly adapt workflows in response to evolving business needs or localized requirements.

By committing to CD organizations can unlock higher efficiency in the Performance and Agility quadrants, enabling workers to focus on higher added value areas in the Innovation quadrant or in the Reliability quadrant. Tasks in these quadrants also require higher creativity and are perceived as more important compared to the other two quadrants. This reallocation of focus can enhance employees' intrinsic motivation and sense of purpose, as described by Self-Determination Theory (Deci & Ryan, 2000).

2.5 Challenges in organizational adoption

Although CD has a compelling value proposition for organizations, challenges regarding adoption and execution might arise as with any new technology. According to Hoogsteen and Borgman (Hoogsteen & Borgman, 2022), while the digital fluency of newer generations entering the workforce enhances organizational readiness for adopting emerging technologies such as LCNC platforms, the organizational integration and governance of Citizen Development (CD) initiatives remain limited. The authors dissect three key areas that organizations need to address to effectively utilize these resources, using a modified TOE (Technology-Organization-Environment) framework. In their adaptation, they emphasize cross-functional collaboration and inter-organizational alignment while removing traditional external environmental factors to more precisely examine the intraorganizational adoption of Citizen Development. In this chapter, we will examine adoption challenges through this framework.

In the technological context, the TOE framework traditionally focuses on analyzing the factors that influence the successful adoption of externally available or already available technologies within an organization (Tornatzky & Fleischer, 1990). However, in the case of Citizen Development (CD), this perspective can be narrowed. CD initiatives typically originate from business units rather than IT departments; therefore, the emphasis shifts from technology acquisition to technology utilization, focusing on how effectively existing LCNC platforms can be leveraged by non-technical users to create value-driven solutions. The position of CD within an organization's process strategy is complementary and often subordinate to larger enterprise information systems (EISs), such as ERP suites or collaborative workspace environments. In conclusion, adoption concerns relate less to new technology

purchases and more to the effective use of already available resources. This view aligns with the findings of MIT CDB & Capgemini Consulting (2011), who emphasize that successful digital transformation (DT) does not primarily result from introducing new technologies but from re-envisioning business processes and models to better exploit existing technological capabilities.

Consequently, LCNC platforms natively embedded in EISs, such as Microsoft's Power Platform or the LCNC platforms of Service Now and Salesforce possess a strategic advantage in adoption, especially in organizations that did not yet decide on initiating CD programs. This observation aligns with Gartner's Magic Quadrant ratings from recent years, which consistently rank mentioned platforms among leaders (Gartner, 2019, 2021b, 2023). This availability also affects perceived complexity and organizational context in a positive way.

In addition to the availability of enabling technologies, two further aspects influence technological adoption in CD initiatives: perceived complexity and perceived risk. According to Borgman and Hoogsteen's findings (2022), the perceived complexity of LCNC platforms is relatively low due to their visual user interfaces and drag-and-drop functionalities, which positively affect adoption rates among non-technical users. On the other hand, perceived risks—particularly those related to data security, compliance, and governance—remain significant barriers. These risks often stem from the limited domain knowledge of business professionals in areas such as cybersecurity or system architecture, highlighting the need for IT oversight and structured governance frameworks.

While the perceived technical complexity of LCNC platforms is generally low or manageable for business users, CD projects introduce additional layers of complexity that extend beyond the technical domain. For an initiative to succeed, non-technical employees must possess or develop a combination of business process design, data literacy, and project management skills. However, business professionals whose primary expertise lies in areas such as marketing, finance, or operations can often lack these competencies, which can significantly raise the perceived complexity and limit the scalability of CD initiatives (Project Management Institute, 2021).

This skill gap reinforces the necessity of active IT involvement—not only in providing technical guardrails for data security and compliance, but also in offering strategic governance and developmental support. As CD initiatives typically operate outside traditional IT structures and the role of a citizen developer is not a formalized position, practitioners often lack a clear framework of accountability, supervision, and lifecycle management. Without adequate oversight, CD efforts risk becoming fragmented or introducing compliance vulnerabilities.

From an organizational perspective the authors emphasize the role of top management support (TMS). As previously mentioned, sources also stated that successful DT in most cases comes in a top-down direction, TMS in this case must advocate for the use of already existing resources and new ways of working, relying on a workforce with their main domain knowledge in other areas. This can cause organizational resistance for several reasons, such as the lack of digital literacy and fears of job displacement. These barriers are not unique to LCNC, but are also common in other technological shifts, particularly AI-driven innovation efforts.

Other than challenges in the workforce's digital capabilities, the role of IT in an organization is also a key influence on successful CD adoption. While CD's proposition is to relieve IT development backlogs, there are also new challenges that might surface. The role of IT developers naturally must shift from an executive to a consulting role, while also maintaining critical infrastructure to the business. This new domain brings the need for a new mentality and a closer IT-business relationship than before. Without clear TMS and acceptance from IT, CD initiatives can face resistance and by extension, failure. Hoogsteen et al. recommends a centralized IT platform approach to coordinate CD projects, arguing that this could beneficially impact CD adoption, through a clear interaction between IT and business domains.

From an inter-organizational aspect, a major challenge is vendor lock-in. While CD projects can provide cost-efficient, light-weight solutions to specific business or operational problems, upscaling these projects might bring additional issues in cost management and deployability. As most cloud solutions, including LCNC platforms utilize pay-as-you-go or freemium pricing, deploying these solutions

on a larger scale must come with extensive cost-benefit analysis including the alternative cost of a pro-code application. Vendor lock-in might also affect scalability and life-cycle management.

Hoogsteen and Borgman (2022) emphasize the importance of vendor support as a critical enabler of successful technological adoption. In the context of Low-Code/No-Code (LCNC) platforms, such assistance is typically expected to extend beyond the point of purchase, encompassing post-implementation support. However, a structural limitation often emerges when LCNC capabilities are embedded within broader digital workspace ecosystems—such as Microsoft 365 or Salesforce. In these integrated environments, vendor priorities often center around the overall suite's productivity or collaboration features rather than the specific advancement of Citizen Development practices. As a result, the specialized support, community-building, and targeted knowledge transfer necessary for sustaining CD maturity may receive limited attention compared to standalone LCNC vendors.

2.6 Individual Challenges

As Hoogsteen and Borgman (2022) focus primarily on organizational and technological dimensions of Citizen Development (CD) adoption, it is equally important to consider individual-level motivators that drive practitioner engagement. As of today, there is an existent research gap on the aspects of CD regarding the individual worker. This paper resorts to one research (Elshan et al., 2025) and general theories on motivation (Vroom, 1964).

On the operational level, insufficient motivation can represent a significant barrier to sustained participation in CD initiatives. We will use Vroom's Expectancy Theory (1964) to break down individual motivators. According to Vroom,

$$M(\text{motivation}) = E(\text{Expectancy}) * V(\text{Valence}) * I(\text{Instrumentality})$$

Because the model is multiplicative, motivation collapses if any factor approaches zero. To motivate employees and individual CD practitioners to overcome several challenges, outside their main pool of knowledge, it is important to fulfil all three factors.

Valence – estimated impact is rooted in the value proposition of CD, by raising efficiency and reducing the effort put into high-frequency processes, shifting manual work towards high creativity or high responsibility processes. Elshan et al. (2024) discuss that in most cases “self-appointed” citizen developers start with simpler, lower-impact solutions, often meant to address their own challenges (time saved, reduced error rate, more meaningful work). However, while creating solutions for others can satisfy altruistic motivation, Expectancy and Instrumentality become highly important for citizen developers to engage in larger projects.

Expectancy—the belief that by effort the desired performance is achievable — is enabled by LCNC vendors (ease of use) and IT support (training, templates, coaching) in the organization. On the other hand, initiating and leading CD projects takes a high level of cross-domain knowledge and skills that are usually developed during the project execution. This natural evolution of hard and soft skills, necessary to navigate both technical complexity (LCNC) and project complexity requires a culture that promotes innovation and continuous improvement. To help individual citizen developers track their impact it is important to provide standardized KPIs and promote success, as high impact projects can often pose risk to the individual, questioning their ability to succeed. (Elshan et al., 2024)

Instrumentality — the belief that delivering CD projects will lead to valued outcomes for the employee (e.g., recognition, appraisal credit, time allocation) (Vroom, 1964). Because a citizen developer is usually a persona rather than a job title, contributions are often “on top” of formal duties (Elshan et al., 2025), which risks misalignment with compensation and role expectations. To foster an impactful CD initiative, the organization should institutionalize rewards that make CD contributions visible, career-relevant, and time-accounted. This also poses a challenge as in most cases there is no clearly defined career path that citizen developers can pursue.

2.7 Literature Summary

As there is constant external and internal pressure on companies for innovation to keep up with dynamic markets and evolving customer expectations, digital resources with lower access barriers provide strategic potential to create differentiating value (Bharadwaj et al., 2013; Marston et al., 2011). To utilize these digital resources, traditional organizations with a centralized IT function (central IT) must shift their focus across all levels of the organization and re-envision their ways of working (Westerman et al., 2011). However, this scale of transformation would likely face major constraints (resource scarcity, IT-business misalignment) if executed fully by central IT (Klotz et al., 2019).

Citizen development (CD) leverages low-code/no-code (LCNC) platforms to harness non-IT professionals' domain expertise by lowering the technical skill requirements to build digital solutions for the processes they manage. This approach is meant to fill the delivery gap between business units' own innovation needs and the limited capacity of central IT-owned resources and professional developers (Verma & Shree, 2025; Hoogsteen & Borgman, 2022).

As this democratization of development can provide a vast amount of additional resources for organizations, the BPM Context Matrix can be realized at lower levels of the organization by potentially reducing the cost of BPM and raising its efficiency thanks to local iterations (see Figure 2). This way "the long tail of business processes" can be optimized, freeing up resources from a local organization's processes in the "Agility" (high variability, high frequency—thanks to hyper-agile development) and "Performance" (low variability, high frequency) quadrants. This reallocation of human effort can be reinvested on a local level in the "Innovation" and "Reliability" quadrants, providing a higher sense of ownership for the workers and a higher level of intrinsic motivation (vom Brocke et al., 2021; Imgrund et al., 2017; Ryan & Deci, 2000).

From an inter-organizational perspective, the importance of understanding the chosen LCNC platform's capabilities and limitations is essential. As LCNC solutions are highly dependent on the platform they are built on, vendor support from the platform providers is critical for successful adoption (Hoogsteen & Borgman, 2022). As of today, there is a lack of literature on comparing EIS-integrated and LCNC-specific vendor platforms. While LCNC specific vendors usually provide more specialized customer service for the clients, EIS integrated platforms might have a subsidiary position compared to the whole ecosystem. However, as users have easier access to solutions already available at an organization's technology pool, EIS platforms can positively impact adoption as per the lower technical complexity of connecting to them (SSO, gaining access, data integration).

From a technological perspective, organizational adoption is strongly influenced by perceived technical complexity (e.g., the ease of building a solution). Even though the value proposition of LCNC is low technical skill requirements, several additional layers of complexity can surface during a CD project, such as data management, UI/UX design, business process analysis, and project complexity. Another technological aspect influencing adoption is perceived risk. As citizen developers often lack formal IT expertise, their solutions can pose significant risks to data security, compliance and governance (Hoogsteen & Borgman, 2022; PMI, 2021).

To tackle these technological adoption challenges (inter-organizational, technological), active IT involvement in CD initiatives is necessary. (1) As choosing the right platform and maintaining its usability requires a deep domain knowledge, that is not expected of business users (citizen developers), IT must take on platform governance role. (2) As LCNC tools are meant to be used by citizen developers, additional complexity layers might obstruct the success of CD projects. (3) Even though the technical knowledge barrier of LCNC development is constantly decreasing (generative AI assistance, younger generations of digital natives entering the workforce) to establish a user-level knowledge base is critical for business users to initiate bottom-up innovation projects. (4) IT involvement is also critical in evaluating and mitigating the risks posed by the solutions developed and in establishing guardrails (often in the platform itself) and security/governance frameworks. To collect these functions, setting up a dedicated CD Competency Center is recommended, that connects the necessary level of support (training, project and development support) and control functions (risk assessment, establishing guardrails and secure dataflow), but also leaves space for business-driven innovation. (Hoogsteen & Borgman, 2022; PMI, 2021)

From an organizational perspective, TMS is critical for establishing a culture of innovation and continuous improvement. While CD provides a new set of tools for business units, business users must step out of their core domain. This requires providing the necessary upskilling opportunities and support to prevent project failures and the erosion of trust in CD solutions. IT must establish a new mindset toward business units from executor to a coach/consultant role. It is also necessary to make organizational changes that enable clear roles, responsibilities and closer communication with the business units (Westerman et al., 2011; Hoogsteen & Borgman, 2022; Elshan et al., 2024; PMI, 2021).

Because citizen developers are usually in a functional role, rather than in an official job title, understanding CD adoption requires understanding individual motivators and barriers. As LCNC platforms are aimed at shifting manual, routine workflows toward high creativity or high responsibility processes, business users might be motivated to utilize these tools and make a positive change in their workloads (Valence). After gaining experience with the use of the platform (higher Expectancy), they might also develop solutions for their colleagues and business units. This natural development relies on high valence use-cases (the positive impact their solution is estimated to make) and high or developing expectancy (learning by doing) to uphold motivation. However, for citizen developers to be deployable at scale, Instrumentality—the belief that delivering a project leads to valued outcomes (e.g., recognition, appraisal credit, time allocation)—must be reinforced by the organization (Vroom, 1964). As CD projects are often placed on top of their domain responsibilities, citizen developers can risk facing misalignment regarding compensation or career advancement. To address these challenges, organizations should institutionalize rewards that make CD contributions visible, career-relevant, and time-accounted. This also poses a challenge that in most cases there is no clearly defined career path a citizen developer can pursue, while staying in their main domain (e.g., marketing, finance). By providing standardized KPIs and successful projects, citizen developers would also be able to measure their impact and be motivated to initiate higher level projects. (Elshan et al., 2024)

3 Empirical Research

3.1 Qualitative Interviews

3.1.1 Analytic Framework and Limitations

The qualitative interviews examine implications derived from the literature summary through the practical lenses of practitioners. Because the literature synthesizes generalized findings without vendor-specific or real-life details, the interviews assess their practical implications in contemporary LCNC/CD settings. Three participants—active in LCNC development, consulting, and training—were interviewed to provide applied perspectives. The interviews used a semi-structured format to allow expansion of topics identified in the literature and to gain a deeper understanding of practitioners' attitudes and experiences; this also enabled probing and the identification of future research topics. Clients were asked to prepare representative use cases prior to the interview. The interview plan is available in Annex 1.

The research relies on a small expert sample ($n=3$), so findings are exploratory and do not claim saturation. Evidence reflects a context with platform/vendor reliance (primarily Microsoft Power Platform) and participants who currently work in Hungary, although they have an international outlook; this may limit transferability to other regions, industries, or LCNC stacks. Some outcome estimates are self-reported where logs are unavailable (recall bias). Finally, the author's (student's) practitioner role introduces potential researcher bias; mitigations included predefined questions, consistent notetaking, and anonymization. Although platform specificity constrains generalizability, it can facilitate operationalization by offering concrete guardrails, policies, and frameworks that organizations can adapt.

3.1.2 Topics Derived from Literature for Empirical Research

To be able to effectively conduct the empirical research, this paper will now dissect the main research questions to discussable implications that are more explorable during the qualitative interviews. These implications were derived from the literature introduced in the previous section.

RQ1 - How does CD contribute to organizational strategy and digital transformation?

Based on literature regarding digital business strategy in general, the aim of DT is to create differentiating value through the utilization of digital resources across the organization (Bharadwaj et al., 2013). CD's integration into digital business strategy can unlock new resources by reducing the cost and coordination overhead of traditional BPM, so the long tail of business processes can be optimized (Imgrund et al., 2017).

Discussable Topics for Expanding RQ1 (RQ / Organizational – Strategy)

- | | |
|----------------|---|
| RQ1/S1: | What are typical LCNC/CD use cases, and what were their pre-deployment characteristics (e.g., paper-based, email-driven, spreadsheet-mediated)? |
| RQ1/S2: | How are typical LCNC/CD use cases distributed across the BPM Context Matrix? |
| RQ1/S3: | How do the delivery effort and/or costs of LCNC/CD projects compare to those of traditional BPM? |
| RQ1/S4: | Which KPIs are used to evaluate the success of LCNC/CD projects? |
| RQ1/S5: | How are capacity and task focus reallocated after LCNC/CD deployments (e.g., time shifted toward Innovation/Reliability activities)? |
| RQ1/S6: | Can CD effectively bridge the gap between innovation needs and IT backlogs? |

1. Table Topics for Empirical Research - Strategic Factors

RQ2 - What organizational factors enable an efficient CD strategy?

As adoption enablers and prohibitive factors surface differently at each level of an organization, this paper will assess the relevant topics using the same top-down approach used in literature. At the organizational level, according to Hoogsteen and Borgman's (2022) modified TOE framework, adoption factors will be assessed with the following refined questions.

Discussable Topics for Expanding RQ2 (RQ / Organizational – Technical/Organizational/Inter-Organizational)

RQ2/O-T1	Is the perceived complexity of LCNC platforms low enough for an average business user (basic digital literacy, e.g., using MS 365 application on a day-to-day basis) to ideate and develop low-code solutions?
RQ2/O-T2	Can the perceived technical complexity of LCNC solutions be reduced by generative AI assistance or publicly available resources (e.g., YouTube channel videos, online forums)?
RQ2/O-T3	How common are end-user initiated projects? What were the outcomes?
RQ2/O-T4	What are the non-technical complexities a citizen developer can face during a CD project/LCNC development and what are the common methods to overcome them?
RQ2/O-T5	What are the perceived risks of CD initiatives by organizations?
RQ2/O-T6	How can organizations mitigate these risks?
RQ2/O-O1	What are common cultural barriers that CD initiatives face in early adoption?
RQ2/O-O2	How important is TMS (top-management support) in overcoming these challenges?
RQ2/O-O3	What is the role of IT in CD adoption? (platform governance, project support, technical support/training, establishing guardrails, risk assessment/management)
RQ2/O-O4	What organizational frameworks (e.g., Competency Center) are usually available that support citizen developers?
RQ2/O-O5	Do organizations provide upskilling opportunities for business users/citizen developers?
RQ2/O-IO1	How do organizations choose the right LCNC platform for CD initiatives?
RQ2/O-IO2	What are the benefits/trade-offs of using an EIS-integrated vs. a standalone LCNC platform?
RQ2/O-IO3	Do LCNC vendors provide sufficient customer service for organizations?
RQ2/O-IO4	What are potential risks of LCNC platforms domain-specificity? Can they be mitigated?

2. Table Topics for Empirical Research - TOE Factors

RQ2 - What individual factors enable an efficient CD strategy?

To further assess individual motivation as an adoption factor RQ2 will be further dissected based on the Expectancy Theory framework (Vroom, 1964).

Discussable Topics for Expanding RQ2 (RQ / Individual – Valence/Expectancy/Instrumentality)

RQ2/I-V1	What are common personal motivators for business users to start CD initiatives (e.g., time saved, fewer errors, helping colleagues, meaningful work)?
RQ2/I-V2	Where have CD solutions freed capacity, and how was that time reinvested (e.g., Innovation/Reliability tasks)?
RQ2/I-E1	What makes business users confident in their abilities to develop LCNC solutions?
RQ2/I-E2	What are common ways citizen developers develop their technical and non-technical skills?
RQ2/I-I1	What are common ways organizations show appreciation toward citizen developers?
RQ2/I-I2	Do you know about organizations who actively promote CD successes, and how does this affect willingness to take on new initiatives?
RQ2/I-I3	What are typical career paths for citizen developers (within their domain or in IT/LCNC specific roles)?

3. Table Topics for Empirical Research - Individual Motivation

3.1.3 Participant Profiles

3.1.3.1 Participant A

Participant A works as a Power Platform consultant and trainer. They hold Microsoft Business Applications MVP status and are a Microsoft Certified Trainer for Power Platform. They founded the Microsoft Power Platform User Group Hungary, a community for local practitioners. Their first experience with LCNC was in 2018, when they were asked to automate HR and learning management processes for about 500 employees without prior low-code experience.

3.1.3.2 Participant B

Participant B leads a low-code development department at a Big Four consulting firm in Budapest. They have a business background and no formal IT training. Participant B joined their current employer in the late 2010s, when low-code tools, especially RPA, started to spread more widely. Although citizen development is not their main focus, their work with low-code projects and organizational governance provides relevant input for this thesis.

3.1.3.3 Participant C

Participant C is a Microsoft Power Platform developer and BI analyst and currently heads the platform function at a global toy manufacturer. Besides this role, they create educational content on LCNC and digital productivity for Hungarian audiences. Their work combines internal platform development with experience in explaining LCNC concepts to non-technical users.

3.1.4 Ethics and Access

All three subjects were recruited by contacting the Power Platform User Group Hungary and then reaching out via LinkedIn. For one participant, I obtained HR approval from the employer via email before the interview. The participation was voluntary without compensation. Interviews were conducted in Hungarian via Microsoft Teams and recorded using the platform's built-in tools. Translations used in the research were produced by the author. Interviews were scheduled for approximately 30 minutes, with flexibility depending on topic depth.

Participant and employer names and any direct identifiers are pseudonymized; client names are anonymized. The pseudonym key is stored separately from study materials on encrypted storage; only anonymized excerpts appear in the thesis/appendix. Interview recordings were optional and agreed before starting. Audio files, raw notes, and the pseudonym key will be deleted after thesis acceptance; anonymized notes may be retained for academic archiving. No client-sensitive information was requested. Participation was consensual and voluntary.

Because this research also includes a real-life case study led by the author (student), case details were not disclosed to interviewees, and interviewees were not asked about that solution. This prevents data/compliance breaches and mitigates conflicts of interest; interview insights were used only for the general LCNC/CD themes specified in the guide.

3.2 Case Study

To assess practical viability of CD further, the author (student) decided to also introduce a real-life case study that took place in a Global Life Science Company's Hungarian Division, further referenced as Life Science Co (name of the company was anonymized for this paper). The case study is describing a bottom-up CD project, which was initiated to streamline cross-functional commercial processes. As the case study describes a project led by the author, this part of the empirical research will not be utilized to evaluate individual motivators behind CD projects, as it would introduce a high risk of researcher bias.

3.2.1 Business Context

Life Science Co is the Hungarian subsidiary of a global company manufacturing OTC medicinal, nutritional and cosmeceutical products available in Hungarian pharmacies. The Hungarian market's pharmacy channel consists of ~3800 pharmacies (according to Life Science Co CRM database) with highly different commercial opportunities ranging from small family pharmacies to large chain pharmacies with global merchandising and POP (point of purchase) marketing standards. The role of

trade marketing is to establish merchandising standards for product visibility in this channel and if necessary, contract pharmacies for shelf-rent agreements and check execution. These activities were planned and delegated by the trade marketing team, then executed by field sales representatives. Payments for the pharmacies were handled through an external agency outsourcing the procurement processes.

3.2.1.1 Merchandising Standards

While in theory, the trade marketing team determined merchandising standards the sales team often gave the feedback that this level of visibility is not possible to achieve in current market conditions. While shelf-rent agreements were meant to boost visibility, in many cases they were not executed as intended especially in independent pharmacies whose shelf space (planogram) was determined by the local owner/manager. Shelf rent contracts did contain execution criteria, but its form (cm/brand) was not utilizable by field representatives who also had other responsibilities often included in their bonuses/compensations in a more emphasized way.

3.2.1.2 Evaluation

While Life Science Co's CRM system did contain a photo-checker module, the trade team did not have an admin interface to view the photos taken by the sales representatives and provide feedback. Theoretically, the trade marketing team was able to evaluate pictures taken by the sales reps at the end of year after asking the CRM contact to export the pictures and look at them one-by-one recording the evaluation in an excel spreadsheet. As this process required a high amount of manual work, these evaluations were often missed, leading to approving the execution of shelf rent contracts without actual evaluation.

3.2.1.3 Contract Management

Shelf rent contracts were written using paper templates that were filled out by the sales representatives when they were visiting the pharmacies. The paper templates had to be filled out in 3 copies and were delivered to the office whenever the sales reps had visited for other purposes. After receiving the contract, a junior/intern employee had to get the contracts signed by Life Science Co and the external agency, responsible for the payments. After receiving the fully signed contracts, the junior employee digitized the contracts by entering the contract details into a spreadsheet and scanning the document. A critical weakness of this process was the physical delivery of the process. Sales representatives often just handed down the contracts for whomever they met, often causing a delay in the signature and digitization process. Contracts were in some cases even lost, resulting in identifying the error upon end-of-year payment. Manual digitization of the contract details was also a challenge rooted in multiple data-entry and unassigned time and compensation for these tasks. It is also important to mention that these tasks were assigned to interns with limited professional experience and common fluctuation in their position.

3.2.1.4 Year-end Payments

Payments were made through an external agency. After year-end evaluation of the contract delivery, an updated spreadsheet was sent to the agency who performed the payments to the pharmacies and validated invoices.

3.2.2 Project Initiation

When the Trade Marketing intern left and no clear owner was assigned for compliance and invoicing visibility, previously known issues began to surface more acutely, creating growing dissatisfaction among sales representatives and pharmacies. Following a sales meeting where these concerns were raised, the Head of Sales commissioned a remediation project.

At that time, stakeholders widely attributed the difficulties to the external agency's service quality. The initial solution proposed was to replace the recurring, agency-mediated invoicing with a "package discount" contract option for pharmacies. Under this proposal, only a small subset of pharmacies (fewer than ten) would have remained on fixed fee compensation, which Life-Sciences Co. HU could process internally without the agency.

Independently, a restructuring in the Trade Marketing function reassigned end-to-end process ownership to a full-time Junior Trade Coordinator ("Trade Coordinator"). Upon taking ownership, the Trade Coordinator reviewed the AS-IS processes, mapped the pain points and prepared a cost analysis for the package-discount option.

3.2.3 Detailed AS-IS Processes and Pain-Points

3.2.3.1 Contract Management Process

To conclude a contract with a pharmacy a sales representative had to pre-approve it with trade marketing for budget allocation. After approval, the sales representatives filled out 3 copies of a paper-based contract template, containing the details of the service (shelf-rent parameters) and the details necessary for the contacts. The contracts were delivered to the office by the sales representatives, usually during the regular company meetings. After receiving the contracts, the trade marketing intern (or junior employee, further referenced as intern) would ask Life-Science Co–HU's representatives for their signatures. After the copies of the contract were signed, they were mailed (by courier) to the external agency for further signatures. The external agency signed the contracts, then sent them back to the Life-Science Co–HU's office, where the Intern digitized them and updated the data to the database (spreadsheet) and the legacy contract life-cycle management system (legacy CLMS). Finally, one copy of the contracts were posted back to the pharmacies.

Challenges

Pre-approval (Sales Rep → Trade Marketing) (1) Contract pre-approvals were informal and usually performed by the managers. Interns were not formally notified before receiving the contracts, so budget allocation was not performed before receiving the physical copies. (2) Shelf-rent criteria for the contract were not fixed early on, and contract formats were not able to provide clear rules for compliance checks.

Prepare 3 paper copies in the field (Sales Rep) (3) Contract details had to be input 3 times on the field which led to errors in the contracts.

Delivering copies to the office (via regular meetings) (4) Field to office delivery usually depended on the sales representatives' office visits, which in some cases led to up to two months' delay. (5) Sales reps just handed down the contracts to several employees in the office, which led to further delays and even lost copies in some cases.

Collect internal signatures (Intern → Company representatives) Handing contracts to several employees created delays and risk of misplaced copies before signatures could be collected. Courier to external agency for signatures (6) Contracts sent to the agency were unsupervised and, in some cases, lost upon delivery, with no feedback.

Digitize and update spreadsheet + legacy CLMS (Intern) (7) Digitizing the contracts was often the first occasion errors in data-input were discovered. (8) Data is entered five times during the whole process – three copies of the paper contract, one time for the spreadsheet and one time to the company's legacy CLMS system. (9) If a contract got lost/delayed and it did not get recorded in the spreadsheet Operating model and capacity (10) Tasks were undocumented and routinely assigned to interns with limited professional experience, resulting in inconsistent handling and knowledge loss. (11) Digitization and data-entry work were excluded from performance evaluation and lacked protected time, causing deprioritization, delays, and error carryover.

Collectively, these issues extended cycle times and increased error rates, while consuming significant capacity for error and exception handling. Beyond internal inefficiencies, they negatively influenced partner relations and introduced exposure to compliance and regulatory incidents.

For further process improvements, three root causes were identified in the contract management process. (1) Paper-based contracts, as the reasons for multiple data entry and delivery issues. (2) Informal, insufficient communication between actors. (3) Lack of resources/incentives for maintaining manual workflows.

3.2.3.2 Compliance Check Processes

Compliance check processes were conducted at the end of each year (Q3-Q4). Sales reps were taking photos of the contracted pharmacies through their CRM system's built-in functionalities, then the images were exported from the CRM to a shared folder, where the trade team could access them. The trade team went through the folder and recorded pharmacies that were insufficiently delivering the service. After updating the spreadsheet database with payable statuses, the payable pharmacies were sent to the external agency.

Challenges

Compliance checks (annual Q3–Q4) Compliance checks were concentrated at year end. Because the trade team lacked real-time access to photos, images were exported from the CRM in bulk, leaving too little time to review all pictures before the payment deadline. Insufficient CRM capabilities (no built-in approvals). (2) With no admin application for trade marketing, the CRM lacked approval functionality, so evaluations were handled manually (e.g., sending feedback by email, moving pictures between folders, and recording results in spreadsheets), leading to late responses and missed follow-ups. (3) The fact of having a contract was not recorded in the CRM system and sales representatives often did not know which pharmacies they should record.

Determining placement criteria (4) Recorded placement criteria depended on outputs from the contract process, so output quality constrained this process. Because comparison rules were not captured in a recordable way on paper templates, compliance decisions relied heavily on the approver's judgment. This led to compromises: (a) providing feedback and requesting improvements for each pharmacy would have required excessive resources; and (b) sales reps were reluctant to jeopardize pharmacy relationships.

3.2.3.3 Invoicing

After the external agency received the list of payable pharmacies, it first verified the accuracy and completeness of the data. In cases of missing or incorrect information (compared to the official company registry), the records with errors were sent back to the intern, who then forwarded them to the responsible sales representatives. Once the corrected details were received, the agency issued invoice requests to the pharmacies. If a pharmacy did not submit the invoice within the expected timeframe, a follow-up reminder was sent before the deadline. The payments to the pharmacies were executed through a bank account registered to the agency.

Challenges

Lack of direct communication between the agency and pharmacies (1) Although the agency was responsible for sending invoice requests and processing payments, it did not communicate directly with the pharmacies. All interactions were intermediated by internal stakeholders (the intern and sales representatives). Handling non-compliant invoices (2) In some cases, pharmacies submitted invoices that were not acceptable due to regulatory or formal compliance issues. When the agency declined these faulty invoices, pharmacies could only contact the sales representatives, who generally did not have the professional knowledge to provide an immediate, accurate answer. As a result, sales representatives had to follow up with the intern, who in turn sometimes had to escalate the question back to the agency, creating delays and inefficiencies.

Outdated pharmacy data identified only at invoicing (3) Some pharmacies did not proactively communicate changes in their company data (e.g., address, tax number, bank account details) during the year. Consequently, these discrepancies were only discovered at the invoicing stage, leading to additional correction cycles and administrative overhead

Overall, these challenges resulted in significantly higher effort to maintain indirect communication between the pharmacies and the agency, and they frequently had a negative impact on the relationship between the pharmacies and Life-Science Co.

3.2.4 Identified Root Causes

The analysis revealed five structural root causes behind the recurring issues in contract management, compliance checks, and invoicing:

- (1) Heavy reliance on paper contracts** and spreadsheet-based tracking, leading to repeated data entry, document loss, and long, non-transparent cycle times.
- (2) Informal, undocumented workflows and communication**, resulting in unclear responsibilities, ad hoc approvals, and person-dependent process knowledge.
- (3) Fragmented systems** (CRM, spreadsheets, legacy CLMS, and the external agency), with no single source of truth and limited end-to-end visibility.
- (4) Under-resourced manual work**, concentrated on interns without KPIs, protected time, or clear ownership, causing inconsistent execution and backlogs.
- (5) An elevated risk of compliance issues and strained pharmacy relationships**, as delays, errors, and limited transparency directly affected contract execution, invoicing accuracy, and perceived reliability.

3.2.5 Solution Requirements

As the cost analysis for the package discount option required sales representatives to provide sell-out data, the Trade Coordinator created a SharePoint list to collect pharmacy data and sell-out values. Later, this SharePoint list was extended and utilized as the unified database for visibility contracts.

Since the Trade Coordinator was aware that Microsoft Power Platform (MPP) was available at Life-Science Co., thanks to the enterprise's MS 365 licensing, an experimental development was initiated to build a comprehensive solution. The solution design was guided by the previously introduced requirements:

Fully digitized documents and unified data storage, accessible to both the Sales and the Trade functions.

- a. Unified data storage is established using SharePoint as the central repository. To make this accessible for sales representatives, a user-friendly interface was created in Microsoft Power Apps.
- b. The Trade Marketing team can access the SharePoint lists directly and use them as an administrative interface.

Standardized processes, supported by automations.

- a. Key process steps for sales representatives (such as initiating contracts and capturing photos for compliance checks) are embedded in the Power Apps user interface.
- b. Administrative processes are triggered by changes in SharePoint records and are executed using Microsoft Power Automate.

Integrated systems for all three main processes (Contract Management, Compliance Checks and Invoicing).

- a. All workflows are executed within Life-Science Co.'s Microsoft tenant, providing immediate access for interventions without the need to contact external parties.
- b. Additional SharePoint lists support the downstream workflows for compliance checks and invoicing.
- c. Compliance criteria are embedded in the contract records and remain available throughout the process from contracting the pharmacy to compliance checks.
- d. While contract management and compliance checks could be fully managed using SharePoint and MPP, after the separation from the external agency a limitation emerged: providing an end-to-end

solution for payments would have required deeper integration with the internally used SAP Ariba procurement module, which was beyond the scope of this project.

Redefined contract templates and compensation options.

- a. New, digital contract templates were introduced.
- b. Insertable picture controls and AI-based object detection are utilized to support customized, contract-specific compliance criteria.

3.2.6 Ideation and Project Execution

3.2.6.1 Gaining Access to Developer Resources

While the citizen developer had access to the MS Power Platform, it has become clear after gathering the requirements that for some functionalities premium features were going to be necessary during development. Life Science Co had a global engagement channel for Power Platform related questions, where an inquiry was sent regarding the process to gain access to MS Power Platform Premium.

In the reply, a general description of the process was provided regarding the governance and security measures required to obtain this access. It later became clear that the approval process was not a dedicated citizen developer/low-code authorization pathway, but the same procedure applied to all custom-developed applications. As no contact information was shared and no active support was offered, several challenges emerged during this phase of the project. The most notable was the requirement to request an Enterprise-level Architecture ID for the application. This request was rejected twice without any explanatory comments, and once local IT support finally provided the approver's contact details, the value of the business case was questioned several times. To resolve this issue, top-management support had to be sought, which resulted in an additional delay of approximately one month.

3.2.6.2 Contract Management

Enabled by LCNC (Microsoft Power Platform), the solution's ideation, design, and development were not strictly separated during project execution. As the project's original focus was primarily on insourcing the invoicing process from the external agency (since inefficiencies were initially attributed to low service quality on the agency's side), the first solution concept took the form of a chain of web automations. These automations digitalized the contract management process, as the existing three-party contracts had to be terminated and renewed without the agency. This approach required close coordination with other functions such as Sales and Legal, which could be directly involved in the development process, enabling faster alignment and fewer feedback loops.

For example, when a new contract template had to be created, the company's Legal function actively contributed to the development of Microsoft Word templates that could be populated via automated cloud flows. This was achieved through a collaborative working session, during which the citizen developer and the company lawyer reviewed the contract together and adjusted the wording so that Word-native Content Controls could be applied.

Once the new contract template was created, the recurring challenge of defining usable delivery criteria (shelf standards) re-emerged. While the citizen developer argued that the pre-existing requirements template was not suitable for compliance-check purposes, trade marketing managers were initially unwilling to modify these requirements into a format that could be implemented in the solution. This led to additional delays until the development of the compliance-check component provided an alternative that the trade marketing function considered acceptable. This created an additional challenge, as the citizen developer's primary domain was trade marketing, and the resulting friction further reduced stakeholder support for the project in his immediate team.

3.2.6.3 Compliance Check Process

At the time of the project, a parallel global initiative was launched with the aim of procuring an AI-powered solution for sales representatives to support compliance-check analysis. The proposed product was designed to generate planogram analytics by recognizing products on shelves and evaluating the

results against predefined criteria. Although this solution had the potential to serve as an effective compliance-check mechanism, it soon became evident that budget constraints would prevent not only its purchase but even the execution of pilot tests.

While the product itself was not implemented, the concept of using AI to support compliance checks remained attractive and was perceived as a viable alternative to the existing manual processes. As Microsoft Azure resources were available at Life Science Co, a parallel internal development was initiated to create an AI-assisted photo-checker tool. Over a two-month experimental development period, an AI model was trained in Azure AI Foundry's Custom Vision platform, and an accessible user interface was built for sales representatives. This was followed by a one-month pilot involving regional sales managers, who introduced the solution to their field sales teams for real-life testing.

The demo tool was quickly recognized and appreciated by the sales teams, who provided weekly feedback for UI/UX enhancements. Based on this positive reception, the project was able to regain the support of the trade marketing function, which subsequently approved the integration of the AI-assisted photo checker with the contract management process. In the final design, the outputs of the AI analytics were formally embedded into the contracts, providing an efficient and operationally usable reference point for delivery approvals.

3.2.6.4 Invoicing

As the project's initial focus was on rearranging contracts and introducing new compliance-check methods, the ideation and development of the invoicing process took place at a later stage. Since Life Science Co's contract with the external agency was only terminated at year-end, the agency's experience in determining taxation requirements was leveraged when designing the invoice query letters sent to pharmacies. While the distribution of these query letters could be automated relatively easily, the development of a fully integrated procurement connection encountered a major constraint. Procurement and accounts payable processes were owned by a global service center, and access to the necessary APIs for initiating purchase requests from within the solution was not authorized. This meant that, during the first year after the solution's deployment, these processes had to be carried out manually. For further development, an RPA-based extension was considered to integrate these steps, but this phase of the project had not yet been implemented at the time of recording this case study.

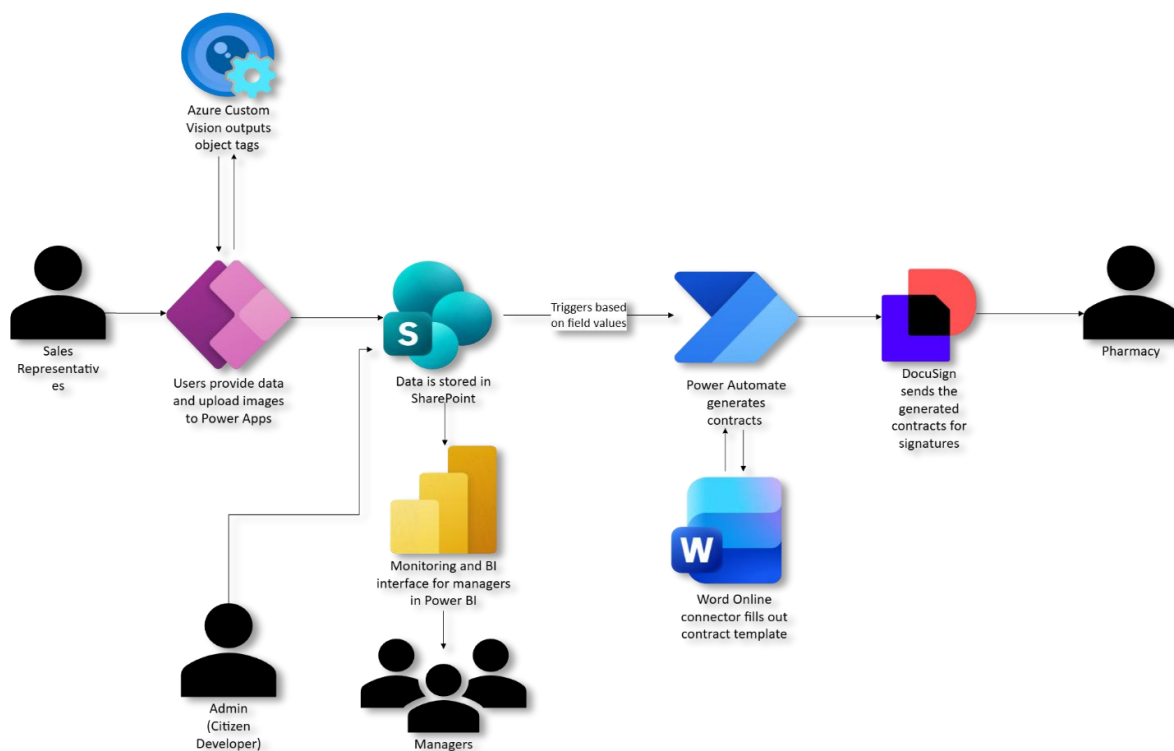
3.2.7 Solution Introduction

This chapter is intended to provide a further practical understanding of the outputs of the project and understand the capabilities and limitations of LCNC solutions (in this case MS Power Platform).

3.2.7.1 Microsoft Power Platform

The Microsoft Power Platform is a vendor-managed cloud service (platform as a service, PaaS) that was formally introduced around 2018. In its initial form, it consisted of three main components: Power BI for business intelligence and analytics, Power Apps for developing low code/no code (LCNC) applications, and Microsoft Power Automate (formally MS Flow) for workflow and web automation. Over time, the platform was expanded with additional services, including Power Automate for Desktop, which added robotic process automation (RPA) capabilities, and Power Pages, which enables the design of externally facing business websites. More recently, Copilot Studio was introduced as a tool for designing autonomous AI agents without coding knowledge; however, unlike the other services, that are usually included in basic MS 365 licenses, Copilot Studio follows a distinct licensing model.

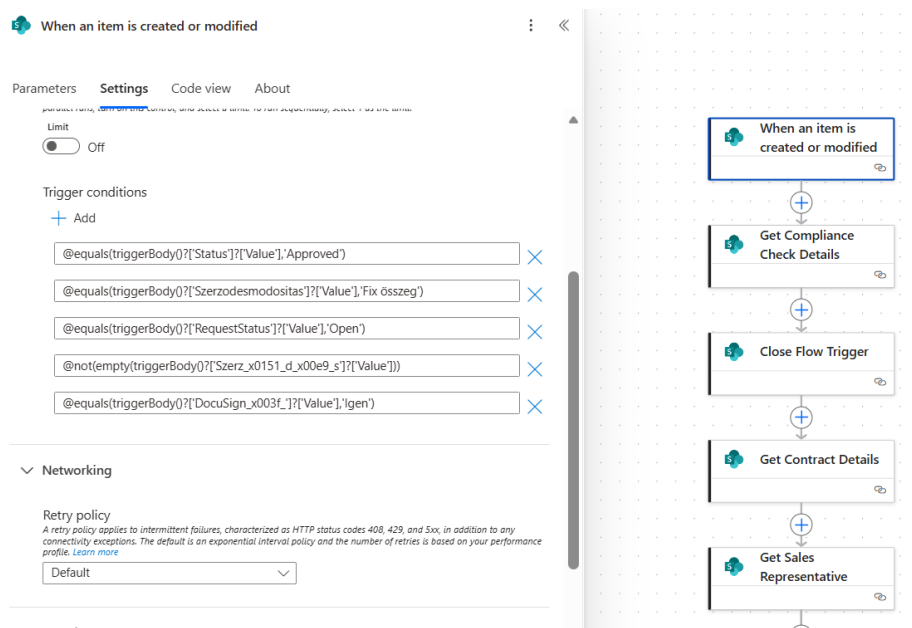
While the Power Platform now offers a broad portfolio of services, this thesis focuses only on those components that are directly connected to the case study and illustrates them with concrete examples of how they were utilized.



3. Figure Solution Overview Diagram

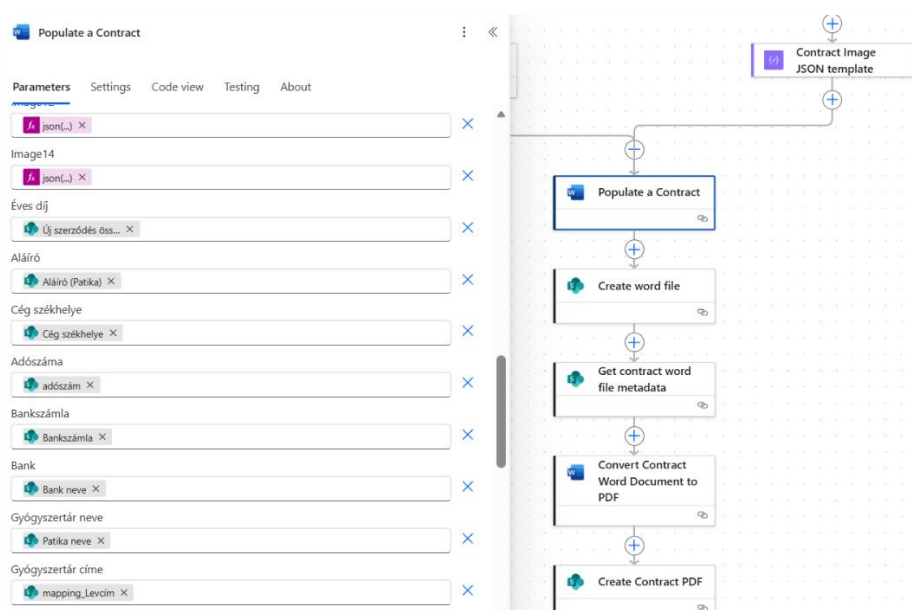
3.2.7.2 Power Automate

Microsoft Power Automate's cloud-based workflow capabilities were used in the project's contract management and invoicing workflows. Automations ("flows") can be started by various events (triggers). In this project, automated cloud flows were configured on SharePoint list events with additional trigger conditions. In practice, a flow was executed only when a list item was created or modified with a specific combination of field values (see Figure 4).



4. Figure Trigger conditions in an automated cloud flow

In the example shown in Figure 4, the trigger conditions ensure that the flow runs only when a contract modification request is submitted in SharePoint with yearly fee-based compensation and the contract is flagged for signature by the pharmacy through DocuSign.



5. Figure Populate the contract template with flow variables

After the trigger conditions are met, the data required to generate the contract is collected from several SharePoint lists, including the pharmacy's contract master data, the objects identified during the visit, and the images taken on site. Once the necessary data has been loaded into flow variables ("Parameters"), the prepared contract template is populated (see Figure 5). After the contract is assembled, the document is forwarded to the partner and Life Science Co's representatives for signature via a third-party integration (DocuSign). Thanks to Microsoft Power Platform's wide range of published connectors, this integration did not require additional custom coding; however, a request had to be submitted to Life Science Co's DocuSign administration team to activate the necessary API connections and webhooks.

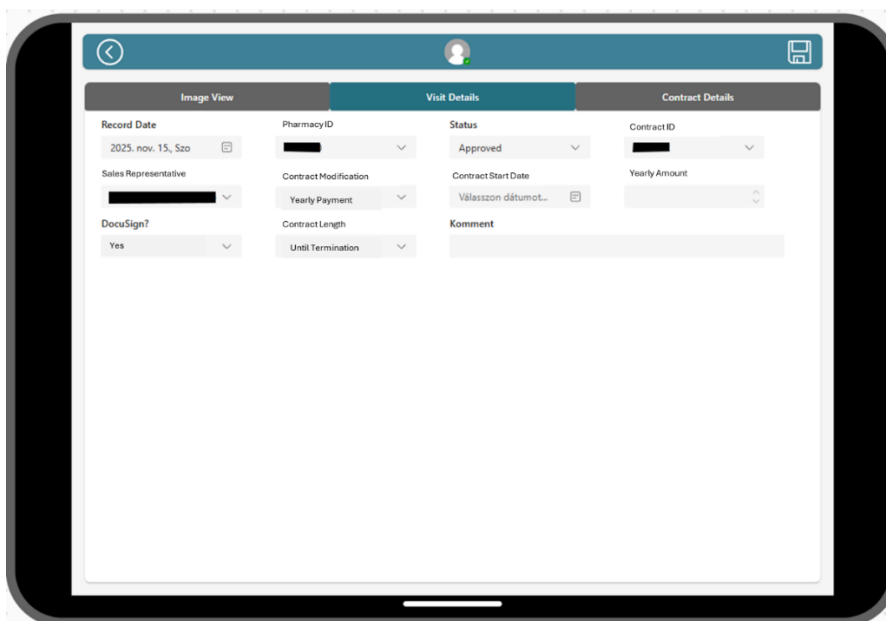
One obstacle faced by the citizen developer was the limitation of the MS Word connector in Power Automate. While plain text (e.g. the product list generated by the AI output) could be inserted dynamically, it was not possible to automatically populate a variable number of images - the action failed if no image value was provided. Because including these images in the contracts was not only technically relevant but also a stakeholder requirement, a workaround had to be developed that introduced custom coding elements. These elements were implemented within 1–2 hours using general-purpose AI assistance, without the need to involve external development resources.

3.2.7.3 Power Apps

Microsoft Power Apps currently offers three primary approaches for creating LCNC business applications. Model-driven apps provide a configurable, grid-like user interface for data stored in Microsoft Dataverse, the Power Platform's native data platform. Canvas apps enable citizen developers to fully customize applications using pre-built elements ("controls") in a drag-and-drop manner, and they can connect to a wide range of data sources beyond Dataverse. More recently, Microsoft introduced Plan Designer, which generates custom solutions based on business documentation and conversational AI prompts.

Although the full capabilities of Power Apps are available when Dataverse is used as the data source, Dataverse follows a separate licensing model that requires all users to have Premium Power Platform licenses. While alternative pricing options exist (such as per-app plans or pay-as-you-go models), the initial difficulties in obtaining premium access led the citizen developer to choose SharePoint as the application's data source and to implement a Canvas app.

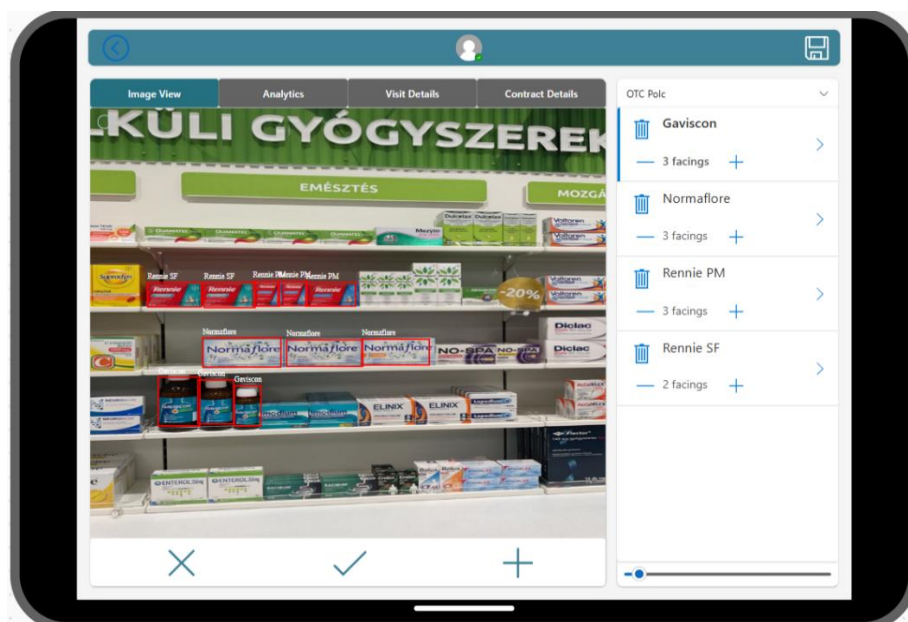
In the case study, the Canvas application serves as a user-friendly interface for sales representatives, enabling them to manage contracts and run AI-based analytics in the field. For the contract management process, a form control was embedded into the app; upon submission, this form writes to a connected SharePoint list, which in turn triggers the corresponding Power Automate flows (see Figure 5).



6. Figure Contract modification request with form control

Contract modification requests had to be submitted with analyzed images, so all related products and images could be inserted into the contract body.

For analyzing the images, the application includes a dedicated photo-capture screen that allows sales representatives to take and upload images directly during their pharmacy visits (see Figure 7).



7. Figure Image upload screen with object detection and correction features

Sales representatives can capture relevant images in the pharmacy or after the visits. The image is then sent to Azure Custom Vision's prediction API that sends back the collection of recognized objects. As the AI model was taught with an experimental/no code approach, the best precision reached until the time of the case study was 76%. To overcome this limitation, the UI was designed so sales representatives are able to edit the list of returned objects as easily as possible.

As Power Apps was not able to dynamically generate the bounding boxes from the relative position data returned by the API, a custom coded SVG component was designed with general-purpose AI assistance.

3.2.7.4 Power BI

To provide analytics for the sales team based on the collected field data, a Power BI dashboard was created and connected to the SharePoint lists. When a pharmacy did not deliver the shelf presence specified in the contract, it was automatically flagged in the report, enabling the trade team to contact the responsible sales representative and take corrective action mid-year, rather than only at year-end before payment approval.

The Power BI dashboard also provided additional insights, such as shelf share and competitor presence. As a further enhancement, the citizen developer began experimenting with linking retail shelf data to sell-out figures, allowing the trade team to assess the effectiveness of their in-store initiatives.

3.2.8 Post-Implementation Challenges

After the deployment of the solution, sales representatives were required to renew all visibility contracts that had previously included the external agency (approximately 80 pharmacies at the time). Overall user feedback from the sales team was positive; however, the contract-renewal wave also revealed reliability issues in the underlying automations.

Of the ~80 Power Automate flow runs, around 10–15 ended in an error state and required manual intervention by the citizen developer. The majority of these failures were traced back to faulty or inconsistent API responses from the third-party integration (DocuSign), rather than to configuration errors in the flows themselves. Although additional checks, feedback loops and compensating steps were added to the flows to handle these cases, a residual error rate persisted.

Although these errors did not derail the project's overall success, they highlighted an important limitation of the LCNC approach: the platform could effectively digitize and orchestrate the core process, but fully robust, mission-critical integrations with external systems still required more advanced, code-based interventions and dedicated IT support.

3.2.9 Impacts

Although not all backlog features had been implemented at the time of recording this case study, the deployed solution effectively addressed most of the previously identified root causes.

First, the reliance on paper contracts and spreadsheet-based tracking was largely removed. All new contracts were created and stored in digital form, and contract data was maintained in unified SharePoint lists. As a result, average contract cycle time decreased from approximately three months to 1–2 weeks, depending primarily on internal signature lead times at Life Science Co. Contract status became monitorable in near real time for both sales representatives and the trade marketing team.

Second, the solution contributed to more formalized workflows and clearer ownership. Pre-deployment training and the introduction of the Power Apps interface made process steps explicit for sales representatives, while updates and outcomes were recorded directly in the underlying SharePoint lists. This reduced dependence on informal communication channels and ensured that responsibilities and process outputs were documented.

Third, the solution reduced system fragmentation for locally owned processes. While a fully end-to-end integration with global procurement systems (e.g. SAP Ariba) was not deliverable within the project scope, all locally controlled activities in contract management and compliance checks were integrated through a unified SharePoint-based data model. This removed the need for duplicate data entry across local spreadsheets and systems, improved data consistency, and provided internal stakeholders with access to a single, consolidated source of contract-related information.

Finally, the automation of document handling and related workflows substantially reduced manual workload. For the affected contract- and procurement-related tasks, the estimated effort per contract decreased from approximately 1.5 hours to around 15–20 minutes, with the remaining time primarily

spent on exceptional cases and residual manual steps (procurement). This reduction directly mitigated the earlier dependence on under-resourced intern roles and supported more timely execution of commercial agreements.

4 Research Findings

4.1 RQ1 – How does CD contribute to organizational strategy and digital transformation?

4.1.1 Interview Findings

While most interviewees did not identify “typical” or standardized LCNC/CD use cases, several common pre-deployment characteristics were observable. These included paper- or spreadsheet-based workflows (e.g., employee skills assessments, manufacturing safety forms, dinner orders for company meetings), complementary processes built around global systems (e.g., tracking purchase requests in Excel alongside SAP), and under-resourced manual tasks.

All three participants emphasized the high customizability of LCNC solutions at a substantially lower cost compared to out-of-the-box or traditionally custom-developed applications. Participant B specifically highlighted SAP automations implemented through RPA as a typical use case that can deliver significant cost advantages compared to developing new components within the global SAP architecture. Participants A and C further argued that, due to this high level of customizability, LCNC solutions can deliver superior quality for local processes, independent of cost considerations. At the same time, the participants agreed that for global workflows or processes with highly standardized outputs, out-of-the-box solutions (such as CRM or ERP systems) generally provide superior quality and robustness.

Across all interviews, CD and LCNC were described as tools that help address demands that would otherwise remain out of the scope of central IT. Participants argued that many local, department-specific use cases, especially smaller automations and workflow tools, would not be automated or digitalized at all without the lower cost, shorter lead times, and higher customizability of LCNC/CD. In this sense, CD was seen as partially bridging the gap between business-side innovation needs and IT backlogs: while core, global processes and system changes still depend on traditional IT projects, LCNC platforms enable operational teams to implement a “long tail” of smaller innovations that would not justify a dedicated IT development initiative.

All participants agreed that LCNC can free up organizational capacity by reducing the amount of manual, repetitive work “where the employees’ creativity does not represent added value” (Participant B). Participant A described a company that was able to indirectly reduce headcount by not replacing certain positions after employees left voluntarily, as LCNC-based solutions absorbed a significant portion of their workload.

Similarly, Participant C argued that in a growing organization (such as the global toy manufacturer where he currently works), citizen development can effectively reduce the costs of scaling up, as the resources of experienced employees can be redeployed to new sites, thereby decreasing the need for additional hires. Participant B also emphasized that LCNC initiatives tend to receive particularly positive feedback in organizations where employees are already overworked, as these solutions alleviate routine burdens and improve employees’ perceived workload.

Participant C explained that LCNC and citizen development projects at his organization are primarily evaluated based on measurable financial impact. The core KPIs include estimated and realized cost savings and “headcount-linked” savings, complemented by an assessment of the development effort required for each application. These metrics are used both ex ante, to decide whether a project is worth pursuing, and ex post, to demonstrate the value delivered by the team. This type of KPI (“headcount-linked savings”) was also considered effective by the other participants, who argued that many of these use cases would not have been automated or digitalized without CD/LCNC’s low cost and customizability.

Participants overall satisfaction with CD programs was divided. While Participant A had a clearly positive opinion, Participant C mentioned that its limitation is the often unclear learning curve and efficiency. Participant B expressed clear concerns regarding governance, architecture and overall viability of the concept (“While theoretically it (CD) can work, I met very few examples where such initiatives were actually able to unlock organizational value.”)

4.1.2 Study Findings

In the Life Science Co. case, citizen development with LCNC was mainly used to fix local commercial processes that were important for daily business but not a priority for global IT. A Trade Coordinator, acting as a citizen developer, used Power Platform to improve contract handling, compliance checks and invoicing around visibility (shelf-rent) agreements. This shows that LCNC/CD is especially suitable for local, cross-functional workflows that are manual, fragmented and hard to manage with traditional tools.

The case also shows that LCNC/CD mainly targets low-creativity, repetitive tasks. Before the project, a lot of time went into typing the same data many times, moving documents between people, and doing late, manual checks. After the LCNC solution was introduced, these basic steps became digital and partly automated. People could then spend more time on follow-up actions, solving special cases and working with pharmacies, instead of just doing administration.

In terms of effort and cost, the solution was delivered by one citizen developer, using tools that were already available in the company's Microsoft 365 environment. There was no need for a separate IT development team or a new external system. The main difficulties were related to governance (access to premium licenses, architecture approval, limits around SAP integration), not to coding itself. This suggests that LCNC/CD can be a faster and cheaper way to improve local processes than a traditional BPM or ERP change, as long as the organization supports the citizen developer and clarifies rules.

The success of the LCNC/CD project in the case was mainly visible through simple operational results. Contract cycle time dropped from about three months to roughly one–two weeks, and the estimated manual effort per contract fell from around 1.5 hours to 15–20 minutes. Data became more consistent and easier to monitor, and issues with pharmacies could be identified earlier. In addition, feedback from sales representatives and fewer complaints from pharmacies indicated higher user and partner satisfaction, even though there was no formal LCNC-specific KPI framework.

After the solution went live, capacity and task focus changed. Routine work such as manual data entry, chasing missing information and correcting basic errors was reduced. The remaining time could be used for exception handling, direct communication with partners and improving the quality of contract execution. Some tasks that were previously given to interns or an external agency could be handled inside the company without increasing workload to the same extent.

4.1.3 Conclusions for RQ1 and Areas for Future Research

These observations are in line with the literature suggesting that CD and LCNC can push down the “line of manageability” in BPM, so that process work and automation can happen closer to the operational level, and people can spend more time on tasks in the creativity and reliability quadrants of the BPM context matrix. In theory, these types of tasks could increase employees' intrinsic motivation, as described by Ryan and Deci's Self-Determination Theory.

However, in this study the interviewees mainly had technology-focused roles, so they were not the best sources for judging how these changes affect employees' motivation in practice. Examining this link more directly, for example by involving non-technical employees, would therefore be a useful direction for future research.

4.2 RQ2 - What organizational and individual factors enable an efficient CD strategy?

4.2.1 RQ2/O

4.2.1.1 Interview findings

Regarding the perceived technical complexity of LCNC platforms, all interviewees agreed that publicly available materials (YouTube videos, forums, free training sessions) provide sufficient knowledge for business users with a basic level of digital literacy¹. According to Participant A, this learning curve can

¹ Using MS 365 applications on a day-to-day basis

be accelerated effectively through organizational support, for example by organizing training sessions in which business users work through a general use case (such as an IT hardware management solution or a coffee machine ordering application). However, as these sessions usually cover one use case, further support and motivation are essential. They also referred to a recent conference discussion, where the future of low-code development was predicted to evolve towards AI-assisted development, thereby removing additional layers of technical complexity for business users (Lamanna, 2025).

Participant B argued that, although the complexity required to create a working solution with LCNC platforms is relatively low, from an IT perspective these solutions often do not comply with established standards, particularly with respect to data-loss prevention. As a possible mitigation, they highlighted the use of tools that cannot access sensitive data—especially RPA platforms—and the creation of managed sandbox environments. However, they also noted that these security layers can be circumvented relatively easily.

Concerning non-technical complexity (such as stakeholder management and project management), all three participants stressed that an effective project strategy can increase the likelihood of success substantially. At the same time, because these skills are not specific to LCNC or citizen development, the interviewees did not view their development as a necessary component of CD training, but rather as a prerequisite for any interdisciplinary initiative.

Overall, while all participants agreed that the technical complexity of LCNC platforms is sufficiently low to enable organizations to initiate CD programs, they were more divided regarding the associated risks that organizations may face.

Participant B highlighted that citizen development can pose a considerable risk to data security, particularly when solutions are built on sensitive datasets such as HR data. They argued that achieving an adequate level of security in a CD solution often requires expertise that exceeds the typical skills of a citizen developer. As a practical example, Participant B described a case in which the underlying data source is imported into a solution without any prior safeguards, and user-relevant records or fields are merely hidden through in-app components. This creates a significant security risk, as the underlying data can be accessed relatively easily by a more experienced user, even if it is not visible in the user interface.

According to Participant A, in most platforms the risk of data leaks can be mitigated effectively by configuring Data Loss Prevention (DLP) policies, for which most LCNC platforms provide built-in capabilities. Drawing on his consulting experience with CD initiatives across several organizations, Participant A suggested that launching such programs from within IT departments and jointly defining governance policies can substantially reduce these perceived risks. Participant C complemented these insights, pointing out that effective data security measures, regardless of CD, usually provide sufficient security for citizen developers (“No access, no building.”). He also noted that his organization applies a citizen development framework that rigorously excludes business-critical processes from CD, ensuring that such tasks remain under traditional IT control. Participant B, in contrast, maintained that built-in security features alone are unlikely to be sufficient to prevent data leaks, and therefore all developed solutions should be monitored individually.

Another limitation, mentioned only by Participant C is CD solutions’ effect on the overall environment’s efficiency. For example, a solution that uses inefficient data-models can slow the overall environments capacity, which can lead to the lagging of more important applications or dashboards. On the other hand, these issues are most likely to surface if the company’s reliance on the LCNC platform is beyond CD initiatives.

Regarding lifecycle management, all participants agreed that this is not an issue specific to LCNC or citizen development, but rather a challenge common to all software development practices. Among the possible solutions, Participant B emphasized the importance of high-quality documentation and thorough stress tests; however, he also noted that citizen developers may lack the capacity or necessary skills to carry out these tasks. Participants A and C, on the other hand, thought that excessively high documentation requirements are not justifiable, as LCNC platforms usually have built-in information features and very high requirements would risk user enablement. Participant C mentioned that a major challenge his organization had to tackle was “orphaned apps”. These are applications owned by

employees who left the company but were not reassigned on the platform. According to this participant, this is a well-known platform limitation of the Microsoft Power Platform and a good example of a possible governance issue.

To overcome these challenges, all participants agreed that a dedicated organizational team should be established to provide guidance and oversight for citizen development initiatives. While such central teams can take various forms (e.g., within IT, as independent units, Centers of Excellence, or community hubs), they were considered essential for creating the organizational culture necessary for CD to succeed.

Participant A described an organization that located its CD center within the IT department and required every citizen development project to undergo the same assessment process as large-scale, custom-coded applications. Although this approach ensured an adequate level of security, the roughly one-month delay caused by these assessments “suffocated innovation by users.”

Participant B suggested that the most promising setup for a CD initiative is a Center of Excellence composed of both IT professionals and experienced non-IT citizen developers, as this structure allows organizations to leverage the benefits of CD—such as bottom-up innovation and rapid development—while still mitigating the associated risks.

Participant C described a similar organizational set-up at his current employer, which he also judged a successful model, although he mentioned that citizen developers on the business side sometimes do not have such a steep learning curve in the LCNC platform, do to comprehensive support that is enabled by the central team.

Participant A also emphasized that the organizational culture around citizen development is often shaped by how the initiative is launched. In his experience, the two most common approaches are (1) top-down and (2) bottom-up.

In the case of top-down initiatives, organizations typically begin by purchasing a number of licenses for an LCNC platform and then organizing some form of training. This approach can be effective, as top management support enables the allocation of resources for necessary organizational changes (such as establishing a Center of Excellence) and training activities. However, because these initiatives are not always grounded in existing business problems, the pressure to demonstrate return on investment may lead business users to develop applications quickly, without having gained the required level of experience.

For bottom-up initiatives, which usually form around an existing challenge faced by a team, a positive return on investment or at least clear added value is often visible from the initial phases, depending on the specific use case. In these situations, however, a much higher level of pressure tends to fall on the individual citizen developer, who frequently lacks sufficient organizational support. Based on examples from all three participants, such projects typically rely on the efforts of a single employee and are therefore at risk of becoming fragmented or isolated within the organization. A common pattern identified during the interviews was that, as more and more solutions emerge within an organization, they raise awareness among both IT and management. At the same time, in these cases there is also a higher risk that the initiative will be deemed unnecessary or overly risky, which can jeopardize the projects.

Participant A highlighted hackathons as a good example of combining top-management support with bottom-up innovation. During these events, companies invite an LCNC consultant or vendor to provide initial training and then support employees in developing solutions, typically framed and financed as a training or professional service.

In conclusion, top-down initiatives are usually more effective in creating the necessary culture and support for CD. While ROI pressure can often lead to frustration, individual citizen developers could set higher goals more confidently, knowing that support is available.

From an interorganizational standpoint, the participants did not identify any major advantages or disadvantages of using EIS-integrated LCNC platforms compared to standalone platforms. However, they

noted that, because most EIS-integrated platforms include a number of “free” licenses as part of a basic subscription (for example, MS 365), they are more likely to be adopted by larger enterprises. Conversely, for smaller organizations, standalone platforms may offer a more cost-effective alternative. Participant A illustrated this by explaining that, for charity organizations, it is often significantly cheaper to develop an LCNC-based fundraising application than to purchase an out-of-the-box software solution.

Domain specificity was addressed by all three professionals as an inherent limitation of LCNC.

Participant A mentioned, for example, a company that built project infrastructure with a planned duration of 20–30 years on the Microsoft Power Platform. Upon follow-up, he did not consider this particularly risky, as the company had prepared migration strategies, and he also argued that for large LCNC vendors (such as Microsoft), vendor lock-in appears less problematic than in the case of smaller providers.

Overall, none of the participants expressed strong concerns about vendor lock-in or other domain-specific risks, although this finding must be interpreted in light of the study’s limitation that the participants’ primary experience was with the Microsoft Power Platform. At the same time, all three participants pointed to weak communication and customer support from LCNC vendors as a recurring issue, which they regarded as an inherent limitation of the current market for which they saw no clear mitigation.

4.2.1.2 Case Study Findings

From a technological perspective, the case study shows that LCNC platforms can meaningfully lower the barrier to building cross-functional business applications, but effective development still requires a hybrid “power user” profile. The citizen developer at Life Science Co. combined SharePoint as a unified data source, a Canvas app in Power Apps as the user interface, automated workflows in Power Automate, an AI model in Azure Custom Vision, and a Power BI dashboard for analytics. Orchestrating these components into an end-to-end process demanded data modelling, understanding of connectors and triggers, and the ability to formalize business logic, indicating that such solutions are feasible for business users with strong digital skills and process ownership rather than for average end users.

The case also illustrates how generative AI and vendor AI services can reduce perceived technical complexity. The AI-powered photo checker was built using Custom Vision’s no-code training interface, and general-purpose AI was used to generate small code snippets to overcome platform limitations (for example, rendering bounding boxes in Power Apps or working around constraints of the Word connector in Power Automate). In this way, AI acted as an on-demand bridge between low code and limited traditional code, extending what a single citizen developer could achieve without involving professional developers.

At the same time, technological risks and constraints became visible around integrations and reliability. Premium platform access followed heavy enterprise architecture workflows, and integration with the global SAP Ariba system was not authorized, limiting the solution’s end-to-end scope. During the first contract-renewal wave, a non-negligible share of Power Automate runs failed due to third-party API responses, and the AI model’s accuracy required a human-in-the-loop design in the user interface. These observations suggest that while LCNC platforms can robustly digitize and coordinate locally owned processes, highly reliable, mission-critical integrations still depend on supportive governance, IT involvement and, in some cases, targeted custom development.

From an organizational perspective, the case study shows that citizen development can take shape even where formal CD frameworks are not yet in place, provided certain supportive conditions exist. At Life Science Co, long-standing practices such as reliance on paper contracts, informal communication and intern-based execution created complexity and ambiguity in process ownership. Challenges initially surfaced through growing operational pain in the field and were first interpreted as issues with the external agency. The situation began to change when these concerns were formally raised and the Head of Sales commissioned a remediation project, which helped reframe the problem as an opportunity to redesign internal workflows and supported the emergence of a citizen-developed solution.

Top-management support also proved important for navigating corporate governance and platform access. Although Power Platform and SharePoint were already available through MS 365, gaining

premium access followed the same architecture approval pathway as fully custom applications and required senior backing to progress. In this context, IT mainly acted as the owner of platforms and integrations and as the guardian of security and compliance standards, while there was no dedicated Center of Excellence or structured citizen development program. Instead, the project advanced through targeted cross-functional collaboration: Legal co-designed automatable contract templates, sales managers piloted and iterated the AI photo checker with their teams, and the trade function gradually adopted revised compliance criteria as the solution demonstrated its usefulness in practice.

Organizational upskilling for CD was largely concentrated in one key role rather than broadly institutionalized. The citizen developer combined deep understanding of trade marketing processes with self-taught LCNC skills, supported by general-purpose AI, and acted as the primary designer and maintainer of the solution. Other business users received focused training on how to use the new application and related workflows, but not on LCNC development itself. This pattern suggests that, even without formal CD structures, organizations can achieve meaningful improvements through motivated “power users” and cross-functional cooperation, while more systematic governance and training would be needed to scale such initiatives more widely and reduce person-dependence.

From an interorganizational perspective, the case study shows that LCNC adoption at Life Science Co was strongly shaped by its existing enterprise ecosystem. Because Microsoft Power Platform was already available through the company’s MS 365 licensing, no separate vendor selection or procurement process was required; the CD initiative could start immediately on an EIS-integrated platform.

The case also highlights that while standard connectors (e.g., to DocuSign, Azure AI services) make it easy to integrate external systems, they introduce a form of dependence on third-party availability and behavior. A non-trivial share of failed Power Automate runs during contract renewal was traced back to inconsistent DocuSign API responses, which local teams could not directly influence and instead had to handle through compensating logic and manual intervention. Vendor and platform communication appeared largely one-directional: a global engagement channel provided general governance guidance for Power Platform Premium access, but there was no dedicated, interactive support model tailored to citizen developers.

4.2.1.3 Conclusions for RQ2/O

Regarding the perceived technical complexity of LCNC platforms, both the qualitative interviews and the case study support the statement that simplicity and ease of use can impact organizational adoption and CD initiatives positively. However, building solutions for complex processes that produce measurable ROI (such as the case at Life Science Co.) typically requires a higher level of digital skills that cannot, in most cases, be expected from average business users. It is also important to note that, with current technological trends (e.g., vibe coding), these entry requirements may be lowered even further.

At the same time, the possibility of lowering the skill barrier further may increase the risks perceived by organizations. These risks are most visible around data security and lifecycle management. While such challenges are not unique to LCNC, citizen developers in most cases do not have the necessary skills and knowledge to address them on their own.

Both data sources point to the need for explicit governance structures and a dedicated CD competency center to balance enablement and risk mitigation. While many governance controls can be implemented using prebuilt functions of LCNC platforms (e.g., data loss prevention policies), the empirical findings suggest that a competency center should extend beyond technical administration. Ideally, it should provide active development support and upskilling opportunities, while also overseeing risk analysis and mitigation.

While bottom-up innovation efforts and CD initiatives can be successful, top-management support is crucial for creating the culture and organizational conditions that can accommodate scalable, more complex CD projects.

From an interorganizational perspective, EIS-integrated platforms can have a positive effect on CD adoption due to their easy accessibility, while interviewees did not identify substantial additional functional benefits. Participants agreed that vendor lock-in is a manageable risk; however, limited vendor support was seen as a factor that may negatively affect adoption.

Overall, citizen development has the potential to become a pillar of organizations' digital business strategy, but top-management support and investments in the necessary organizational frameworks are required to reduce risks and support scalable adoption

4.2.2 RQ2/I2

4.2.2.1 Interview Findings

All interviewees highlighted the positive effects that LCNC and citizen development can have on individual employees, primarily by eliminating manual, repetitive work and enabling a shift towards more creative, higher value-added tasks. For employees to decide to invest time in developing their own solutions, the anticipated impact of the application must be high enough to justify both learning the LCNC platform and navigating the necessary administrative processes (e.g. access requests, approvals). In terms of Vroom's Expectancy Theory, this implies that both Valence (the expected positive impact of the solution on one's own and the team's work) and Expectancy (the belief that one can successfully implement the solution using LCNC tools) need to be sufficiently high. Interviewees also emphasized that many of the most successful CD projects arise in organizations where employees are chronically overworked or understaffed, as the potential efficiency gains and workload relief make the perceived value of CD particularly salient in these contexts.

At the same time, this pattern suggests a potential tension: while high workload increases Valence by making efficiency gains more attractive, it may also risk lowering Expectancy and Instrumentality if employees feel they lack the time, support or recognition to engage in CD sustainably. Participant B even argued that a conceptual weakness of CD is that "workers are doing the job of developers for free." While altruism toward their teams can be a driver of initial engagement, interviewees stressed that for CD initiatives to remain viable, individuals must also perceive a clear and tangible positive impact for themselves (e.g. reduced workload, recognition, or career benefits). Participant C noted that his own entry point into LCNC and CD was a structured training program that explicitly allocated time for these activities (supporting Expectancy) and provided formal recognition upon completion (supporting Instrumentality), which substantially increased his motivation to participate.

Interviewees noted that, in many organizations, CD opens up career paths primarily toward developer or other tech-related roles, rather than strengthening employees' recognition and progression within their original professional field. As a result, some potential citizen developers may hesitate to invest heavily in CD if it appears to pull them away from their core domain. In a few cases, participants mentioned companies that provide explicit financial rewards or other formal recognition for impactful CD projects and ideas while allowing employees to remain in their primary roles. From an Instrumentality perspective, these practices are highly positive: they create a clear line of sight between CD contributions and valued outcomes for the individual, and they align organizational performance goals with the worker's own career interests (the employee stays in their professional domain but is still motivated to help the organization run better). However, interviewees emphasized that such arrangements are still exceptional rather than widespread.

While all interviewees mentioned tech-related positions as a natural progression from a citizen developer role, Participant B mentioned that for moving into a developer or process consultant position, the practical knowledge gained during CD projects is often not enough, because these employees are missing the necessary theoretical foundations, such as reusability, architecture design and data security.

2 Findings are only based on interview findings due to possible researcher bias.

Participant A mentioned that for younger workers, CD participation can be an effective way of drawing attention and recognition in their early careers and often these skills might be reusable during later job applications.

4.2.2.2 Conclusion for RQ2/I

Companies aiming to leverage LCNC and citizen development should deliberately increase the perceived value, feasibility, and personal benefits of CD work by focusing on high-pain repetitive processes, providing time and training to build solutions, and formally recognizing impactful contributions within employees' existing roles. Without these conditions, CD is likely to remain a marginal, one-off activity rather than a scalable, sustainable element of the organization's operating model.

5 Conclusion

The findings show that there are clear examples of citizen development (CD) extending the reach of digital transformation in organizations, primarily due to its low cost and high degree of customizability. Consequently, CD and LCNC solutions can, in certain contexts, substitute out-of-the-box or traditionally custom-developed applications, while delivering comparable or even higher added value. At the same time, for global workflows or processes with highly standardized outputs, out-of-the-box solutions supported by traditional IT development (such as CRM or ERP systems) generally provide superior quality and robustness.

Overall, the empirical insights support the view that CD can effectively bridge the gap between business needs and limited IT resources by addressing the “long tail” of business processes at lower, operational levels of the organization. The outcomes of CD initiatives are most visible in improved employee well-being, headcount-related savings, and higher-quality day-to-day processes.

However, the interviewees were divided regarding the overall success rate and scalability of CD initiatives. To help organizations better harness their potential benefits, this study formulates practical recommendations for establishing a workable CD model. These are grounded in the findings related to RQ2 (“What organizational and individual factors enable an efficient CD strategy?”).

5.1 Recommendations

- **Establish a dedicated CD governance unit.**

Create a separate organizational unit (internal or external) responsible for governing the LCNC platform and overseeing CD projects.

- **Provide initial training and clear frameworks.**

Offer structured introductory training on the chosen LCNC platform and communicate clear frameworks for use, in order to accelerate adoption among business users. For further skills development, AI assistance and publicly available resources (e.g. tutorials, forums) can provide sufficient support during development.

- **Balance governance and accessibility.**

Design governance mechanisms for CD projects and solutions that strike an appropriate balance between risk mitigation for the organization and maintaining the lowest possible entry barriers for citizen developers.

- **Define lean documentation requirements.**

Specify documentation standards that are clear, proportionate, and achievable by business users with minimal time and effort, while still supporting maintainability and lifecycle management.

- **Leverage built-in platform guardrails.**

Configure and utilize the LCNC platform’s built-in guardrails (such as data loss prevention and role-based access controls) in line with the organization’s overall data security policies. Ensure that additional support and oversight are available for CD projects that need to go beyond these default guardrails.

- **Monitor platform-wide impact.**

Use platform-level reporting and monitoring tools to assess CD solutions in terms of the risks they pose to the organization’s overall architecture, performance, and efficiency.

- **Embed experienced CD practitioners.**

Involve experienced citizen developers in the dedicated CD unit and make them accessible across the organization (e.g. as mentors, reviewers or internal consultants).

- **Connect top-management support with bottom-up initiatives.**

Link executive sponsorship to bottom-up CD efforts so that projects and an enabling culture can develop organically rather than remaining isolated. In practice, anchoring CD initiatives in existing business challenges or demonstrably successful projects can increase acceptance and support early ROI.

- **Escalate business-critical solutions to stronger oversight.**

When a citizen-developed solution becomes business-critical, involve the dedicated LCNC platform team and/or the traditional IT department to provide additional oversight and mitigate potential risks.

- **Provide support and recognition for CD upskilling.**

Offer structured upskilling opportunities and, where possible, formal skill certification for citizen developers. This not only increases the overall success rate of CD projects, but also strengthens motivation by raising Expectancy (confidence that success is achievable with the available skills and support) and Instrumentality (the perception that CD efforts are meaningfully aligned with organizational goals and will be acknowledged).

- **Measure success through standardized KPIs and link it to compensation.**

Define clear, standardized KPIs for evaluating CD initiatives (e.g. cost savings, time reduction, quality improvements, or user satisfaction) and connect these, where appropriate, to formal reward mechanisms. This can increase Instrumentality (successful CD work leads to tangible rewards) and Valence (the perceived impact and importance of CD projects). In addition, recognizing CD contributions within employees' primary professional domains can reduce the pressure to move into purely technical roles, allowing them to remain in their core field while still being valued for process and innovation work.

5.2 Possible Areas for Future Research

Future research could focus on how citizen development affects employee motivation and everyday work experience. In this thesis, most interviewees had technology-focused roles, so it is still unclear how non-technical employees feel about CD, how it changes their perception of their job, and whether taking part in development work increases or decreases their engagement and satisfaction.

Another important topic is the role of generative AI and "vibe coding" in LCNC and CD. The findings suggest that these technologies can lower the entry barrier for citizen developers and extend what a single person can build. However, it is not yet clear how this changes the required skill set, how it affects the balance between low-code and traditional coding, and what new risks or dependencies it creates for organizations.

Future research could also look more closely at interorganizational dependencies and vendor ecosystems around LCNC platforms. The thesis highlighted weak vendor support and reliance on EIS-integrated platforms as recurring challenges, but their full impact on reliability, scalability and risk has not been explored in depth.

Finally, there is a need for more comprehensive economic evaluation of CD at enterprise level. While many projects show strong local benefits, it is still an open question how CD influences overall IT costs, IT backlogs, time-to-value of digital initiatives and long-term organizational performance.

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7 Annex

7.1 Annex 1 – Interview Guide

Welcome and introducing GDPR. Consent for recording the interview.

// Recording start

Subject Profile

Please provide a short introduction of your professional background relevant to the research topics (Low Code and Citizen Development).

CD and LCNC in Organizational Strategy and Digital Transformation

[Focus on RQ1]

Could you provide some LCNC use cases you worked on with their main characteristics?

How would you characterize the workflows optimized from a strategic perspective of the organization?

Why were these workflows not optimized before? Why weren't custom solutions built/commercial solutions purchased for these tasks?

[Probes: cost efficiency, customization requirements, delivery effort ,other?]

Were these projects successful? Did you measure KPIs post-implementation? What KPIs did you use?

Did the solution freed up resources? Were these resources reinvested?

[Probes: time/money saved, shift towards more creative/important tasks, employee well-being]

Did the client have an IT department? Where they unable/unavailable to provide a solution for the use case?

[Probes: resources scarcity, business-IT misalignment, misunderstanding business needs]

CD and LCNC adoption

[Focus on RQ2/O-T1-T4]

Did the organization that you worked with have a CD initiative?

[Probes:

If yes: Did they try to build a solution for the use case?

If no: Could these use cases been solved by average business users³ utilizing LCNC solutions?]

What do you think of the technical complexity (or perceived technical complexity) of LCNC platforms? Is it low enough for average business users to develop their own solutions for their use cases?

What do you think of publicly available resources and generative AI assistance? Could these make the development seem more achievable?

[Probe (additional topic, if we have time): Vibe Coding]

What are non-technical complexity layers a citizen developer can face when building a solution?

³ basic digital literacy, e.g., using MS 365 application on a day-to-day basis

[Probes: project management knowledge, compliance, accessibility, lack of sponsorship]

[Focus on RQ2/O-T5-T6]

What are the (risks) organizations face when providing LCNC tools for business users?

[Probes: data-security, enterprise architecture/governance issues, life-cycle management, regulatory/compliance risks]

How can organizations mitigate them?

[Probes: built-in guardrails, spot/suitability assessment, restricted APIs/connectors, other?]

[Focus on RQ2/O-O1-O4]

How do organizations react to the idea of CD? Are there cultural barriers?

Can these barriers be lifted by TMS?

How do IT departments react to CD initiatives? How can they be involved efficiently?

What are the roles IT departments usually play in CD initiatives?

[Probes: platform governance, project support, technical support/training, establishing guardrails, risk assessment/management]

Did you meet any organizational frameworks that supported CD initiatives?

[Probes: Competency Center, CoE, Hub, Community of Knowledge]

[Focus on RQ2/O-O5 & RQ2/I-E1-E2]

What do you think makes citizen developers confident enough to start developing solutions using LCNC? How do they usually develop their skills?

Did you meet companies providing upskilling programs for citizen developers? In what areas?

[Probe: How did **you** learn to use LCNC tools?, training for technical and non-technical skills]

[Focus on RQ2/I-V1-V2]

What factors do you think motivate the start of CD programs?

[Probes: freed capacity, time saved, fewer errors, helping colleagues, more meaningful work]

[Focus on RQ2/I-I1-I3]

In your experience do organizations reward citizen developers? What are common ways?

Do organizations promote CD successes? Do you think it motivates business users to initiate CD projects?

What are typical career paths citizen developers take?

[Probe: inside or outside their main domain, transition to tech roles]

// End of protocol.

Do you have any other insights/topics you think are worth mentioning?

Thank the interviewee and farewell.

7.2 Annex 2 - Ethical statement on the use of GAI

Ethics statement

on the use of a generative artificial intelligence system or service when preparing a thesis, portfolio, or final project.

I, **Marcell Pörczi** (Neptun code: **BHTFYL**), being aware of my ethical responsibility based on the Code of Ethics of Corvinus University of Budapest and in accordance with it, declare that I have completed the thesis related to the completion of the **Business and Management BSc** programme by using a generative artificial intelligence system or service (hereinafter: GAI) as per the following:

(Underline the appropriate item(s) in the list below!)

- 1) I have not used a GAI system or service
- 2) I used a GAI-based system
 - a) In the research and selection of academic literature
system name and version: **ChatGPT 5.1 "Thinking"**
 - b) For preliminary analysis and evaluation of data
system name and version: _____
 - c) For writing, developing and testing computer code
system name and version: _____
 - d) For stylistic and grammatical improvement of textual content, paraphrasing
system name and version: **ChatGPT 5.1 "Thinking"**
 - e) For the development and creation of visual or audiovisual content
system name and version: _____
 - f) To translate certain passages (e.g. between Hungarian and a foreign language)
system name and version: **ChatGPT 5.1 "Thinking"**
 - g) Other:
system name and version: _____

I declare that I have used the new GAI-generated content elements as allowed by the University's GAI regulations and that I have indicated this everywhere in the submitted work the way required by the Thesis Guidelines of my Study Programme (e.g. in a footnote with the note "*GAI-generated content*" and the identification of the scope).

I am aware that the University may use artificial intelligence detection to screen the work I submit and, depending on the outcome, may even initiate an ethics procedure against me.